

The Joy of Pseudo-Boolean Proof Logging

Jakob Nordström

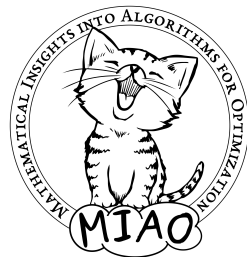
University of Copenhagen and Lund University

ACP Summer School 2026

CP: Explanation, Verification, and Applications

Changchun, China

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The Joy of Pseudo-Boolean Proof Logging

Or: An Introduction to VERIPB

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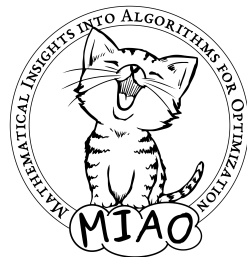
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Brief Presentation

Jakob Nordström

University of Copenhagen
and Lund University

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www.jakobnordstrom.se

Research interests:

- Computational complexity theory
- Applied combinatorial optimization
- Certifying algorithms / proof logging



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Research interests:

- *Proving exponential lower bounds for NP-hard problems*
- *Designing algorithms to solve NP-hard problems in (ideally) linear time*
- *Generating machine-verifiable proofs that such algorithms reason correctly*



Tutorial Based on Research Contributions By...

■ Berhan Oumer Adame

■ Markus Anders

■ Jeremias Berg

■ Bart Bogaerts

■ Benjamin Bogø

■ Wolf De Wulf

■ Emir Demirović

■ Simon Dold

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Combinatorial Solving and Optimisation

- Revolution last couple of decades in **combinatorial solvers** for
 - Boolean satisfiability (SAT) solving [BHvMW21]¹
 - Constraint programming (CP) [RvBW06]
 - Mixed integer linear programming (MIP) [AW13, BR07]
- Solve NP-complete problems (or worse) very successfully in practice!
- **Except solvers are sometimes wrong...** (Even best commercial ones) [BLB10, CKSW13, AGJ⁺18, GSD19, BMN22, GCS23]
- Solvers can propose infeasible “solutions” (but such errors can be detected)
- More challenging: How to achieve reliable claims of infeasibility?
- Or that a solution is optimal? (Even off-by-one mistakes can snowball into large errors if solver used as subroutine)

¹See end of slides for all references with bibliographic details

What Can Be Done About Solver Bugs?

■ Software testing

Very useful, but bugs slip through even with careful domain-specific testing

Progress using [fuzzing](#) and [delta debugging](#) [BB09, BLB10, KB22, NPB22, PB23]

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- **Formal methods**

Prove that solver implementation adheres to formal specification
Current [formal verification](#) techniques cannot scale to level of complexity in modern solvers
(Despite valiant efforts in, e.g., [Fle20])

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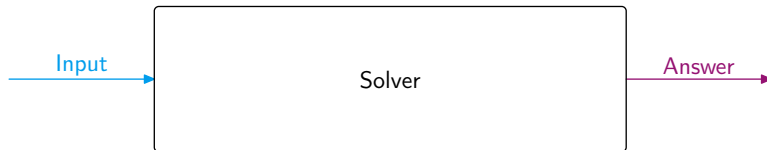
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■ Proof logging

Make solver [certifying](#) [ABM⁺11, MMNS11] by adding code so that it outputs

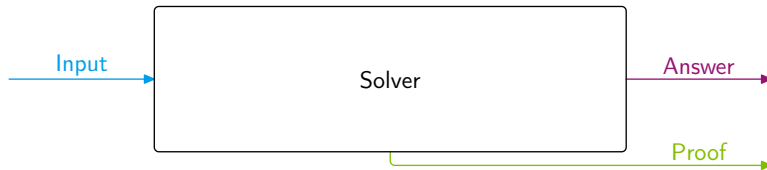
- 1 not only [answer](#) but also
- 2 simple, machine-verifiable [proof](#) that answer is correct

Proof Logging with Certifying Solvers: Workflow



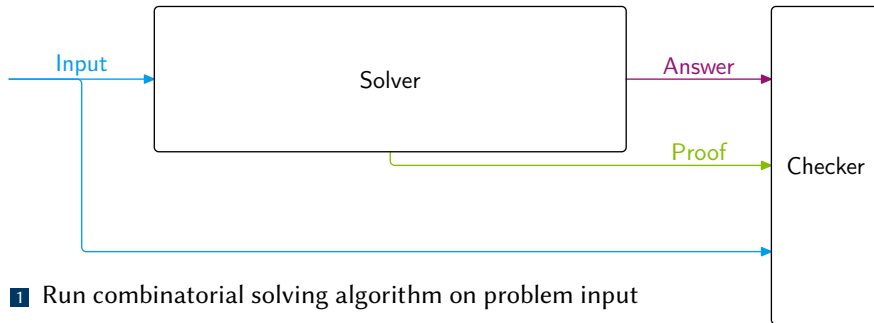
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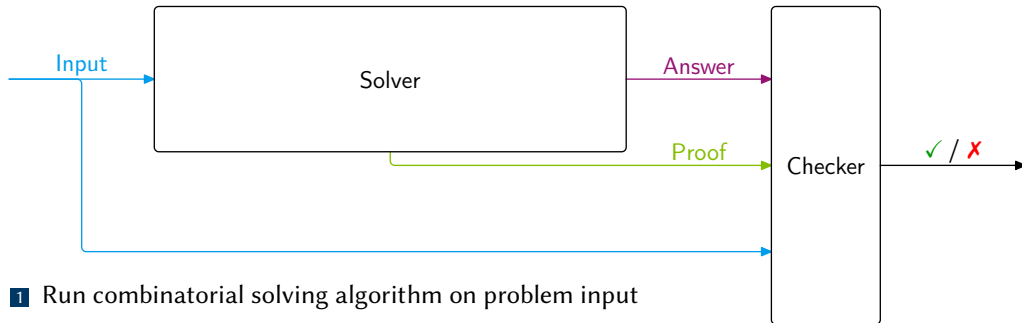
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Proof Logging with Certifying Solvers: Workflow



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- 3 Feed input + answer + proof to proof checker

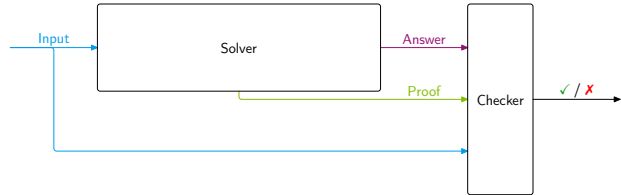
Proof Logging with Certifying Solvers: Workflow



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- 4 Verify that proof checker says answer is correct

Proof Logging Desiderata

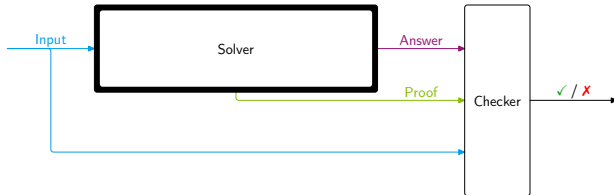
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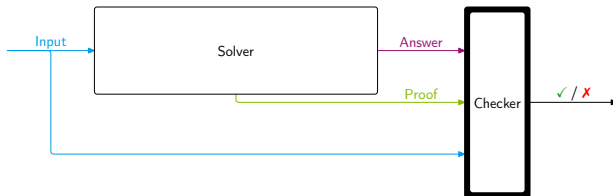
- **very powerful:** minimal overhead for sophisticated reasoning



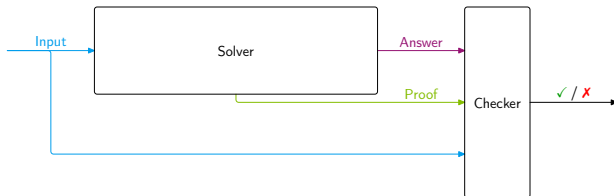
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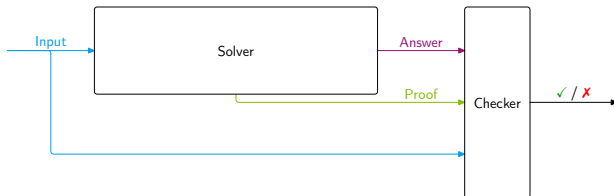


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Clear conflict expressivity vs. simplicity!

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Clear conflict expressivity vs. simplicity!

Asking for both perhaps a little bit too good to be true?

Some Previous Proof Logging Work

Boolean satisfiability (SAT) solving

- Well established since over decade with several proof formats such as [DRAT](#) [HHW13a, HHW13b, WHH14], [GRIT](#) [CMS17], [LRAT](#) [CHH⁺17], ...
- But no efficient support for most advanced techniques such as
 - Gaussian elimination
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- And only for decision problems, not optimization (MaxSAT)

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Mixed integer linear programming

- Work on proof format [VIPR](#) [CGS17, EG23]
- But only for exact solving and without support for advanced techniques

Take-Away Message from This Tutorial

Proof logging for sophisticated combinatorial solvers is possible!



Take-Away Message from This Tutorial

Proof logging for sophisticated combinatorial solvers is possible!

- With **single, unified method!**
- Producing proofs in simple format calculating with **0–1 integer linear inequalities** (a.k.a. **pseudo-Boolean constraints**)
- Implemented in **VeriPB** (<https://VeriPB.org>)



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These slides with references are online at <https://jakobnordstrom.se/presentations/>

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- 7 Serves as stepping stone towards explainability

Design Principles for Proof Logging

Proof logging implementation

- Don't change solver
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- Proof logging overhead small constant fraction of running time ($\lesssim 10\%$)
- Proof checking time within constant factor of solving time (current aim $\lesssim \times 10$)

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Proof system

- Keep language simple — no XOR constraints, CP propagators, symmetry groups, ...
- But reason efficiently about such notions using power of proof system
- Combine proof logging with formally verified proof checker

But Let Us Start from the Beginning...

Review of some basic concepts:

- SATISFIABILITY (SAT) problem
- Unit propagation
- DPLL and CDCL algorithms
- Resolution proof system

The SATISFIABILITY (SAT) Problem

- **Variable** x : takes value **true** ($=1$) or **false** ($=0$)
- **Literal** ℓ : variable x or its negation $\bar{x}$
- **Clause** $C = \ell_1 \vee \dots \vee \ell_k$: disjunction of literals
(Consider as sets, so no repetitions and order irrelevant)
- **Conjunctive normal form (CNF) formula** $F = C_1 \wedge \dots \wedge C_m$: conjunction of clauses

The SAT Problem

Given a CNF formula F , is it satisfiable?

For instance, what about:

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge \\ (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

Proofs for SAT

For satisfiable instances: just specify satisfying assignment

For unsatisfiability: a sequence of clauses (CNF constraints)

- Each clause follows “obviously” from everything we know so far
- Final clause is empty, meaning contradiction (written $\perp$)
- Means original formula must be inconsistent

What Is Obvious? Unit Propagation

Unit Propagation

Clause C **unit propagates** ℓ under partial assignment ρ if ρ falsifies all literals in C except ℓ

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Proof checker should know how to unit propagate until saturation

Davis-Putman-Logemann-Loveland (DPLL)

DPLL [DP60, DLL62]: Assign variables and propagate; backtrack when clause violated

“Proof trace”: when backtracking, write negation of guesses made

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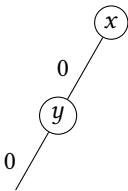


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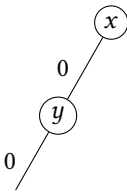


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“Proof trace”: when backtracking, write negation of guesses made

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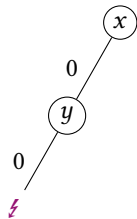
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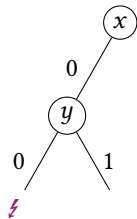
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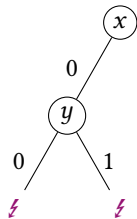
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- 1 $x \vee y$
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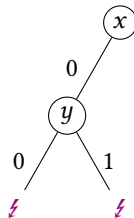
“Proof trace”: when backtracking, write negation of guesses made

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee \textcolor{violet}{x} \vee y) \wedge (\textcolor{violet}{x} \vee \bar{y} \vee z) \wedge (\bar{x} \vee \textcolor{teal}{z}) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

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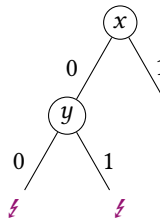
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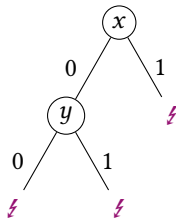
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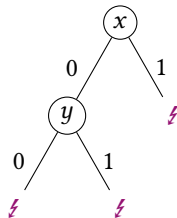
1 $x \vee y$

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3 x

4 $\bar{x}$

5 $\perp$



Reverse Unit Propagation (RUP)

To make this a proof, need backtrack clauses to be easily verifiable

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Fact

Backtrack clauses from DPLL solver generate a RUP proof

What About Conflict-Driven Clause Learning (CDCL)?

Run CDCL [BS97, MS99, MMZ⁺01] on our favourite CNF formula:

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Free choice to assign value to variable

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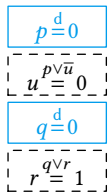
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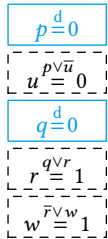
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$$\perp$$

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decision
level 1

decision
level 2

decision
level 3

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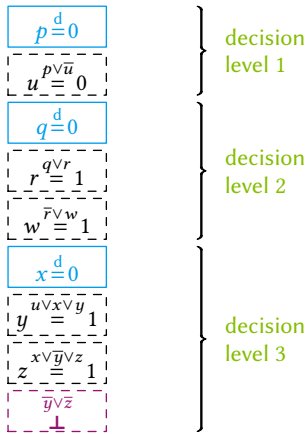
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Conflict Analysis

Time to analyse this conflict and learn from it!

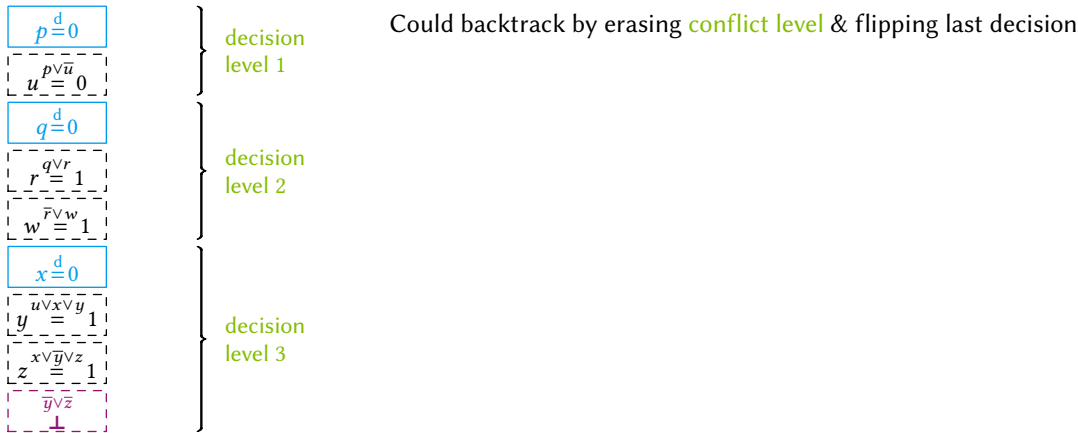
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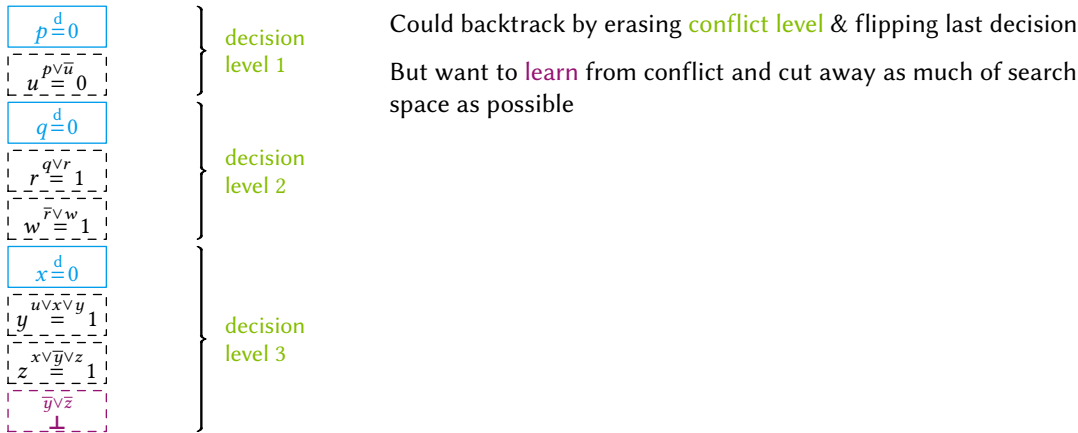
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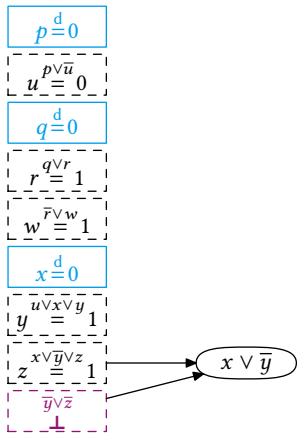
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Could backtrack by erasing **conflict level** & flipping last decision

But want to **learn** from conflict and cut away as much of search space as possible

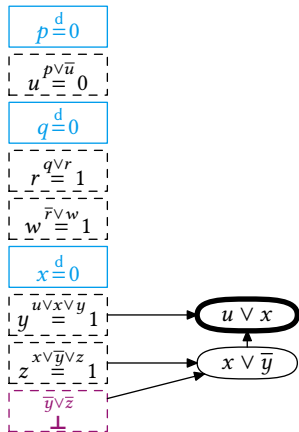
Case analysis over z for last two clauses:

- $x \vee \bar{y} \vee z$ wants $z = 1$
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- **Resolve** clauses by merging them & removing z — must satisfy $x \vee \bar{y}$

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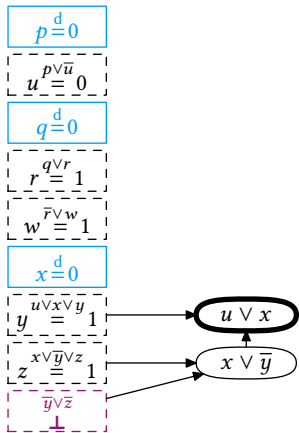
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Repeat until **UIP clause** with only 1 variable at conflict level after last decision — **learn** and **backjump**

Complete Example of CDCL Execution

Backjump: undo max #decisions while learned clause propagates

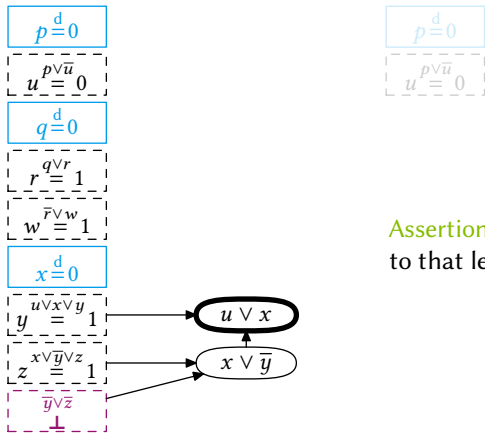
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$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$

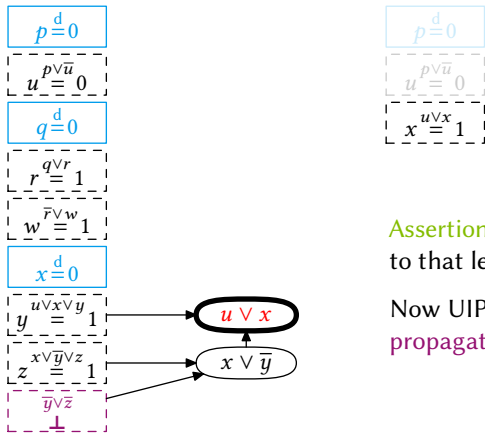


Assertion level 1 (2nd largest level in learned clause) — trim trail to that level

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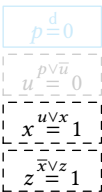
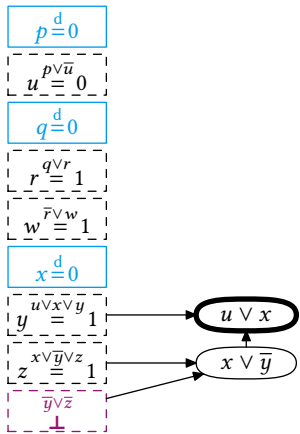
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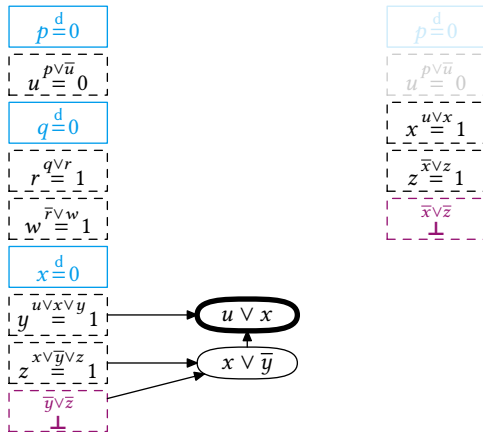
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Then continue as before...

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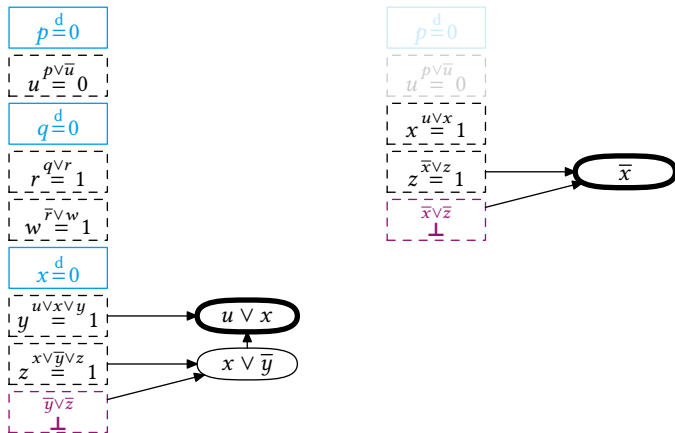
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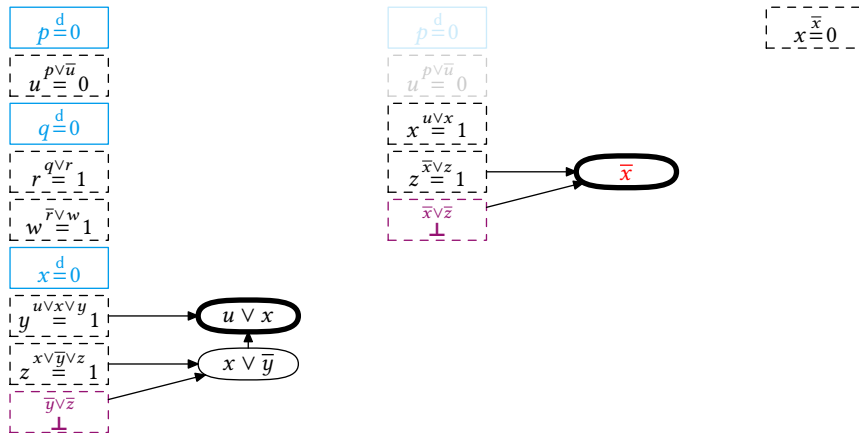
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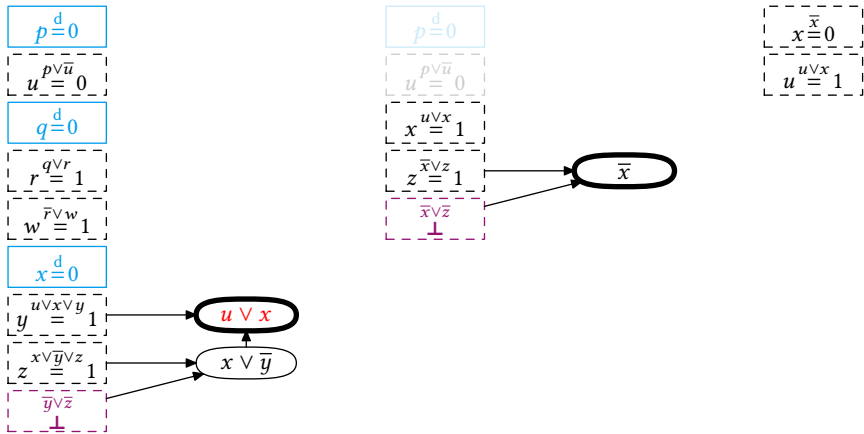
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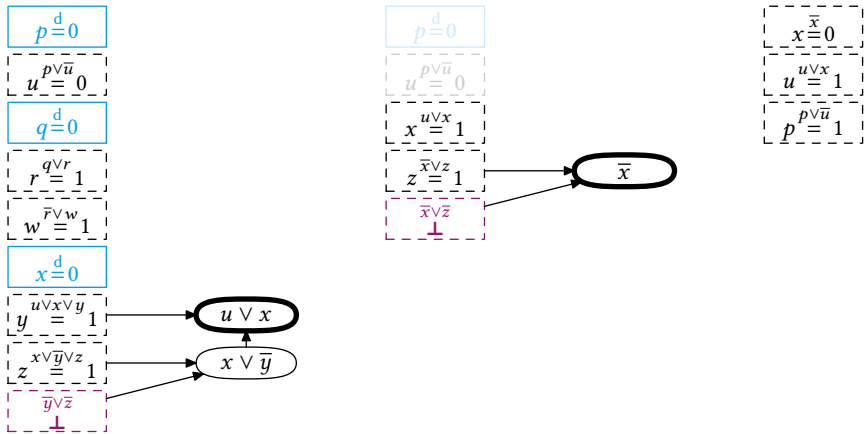
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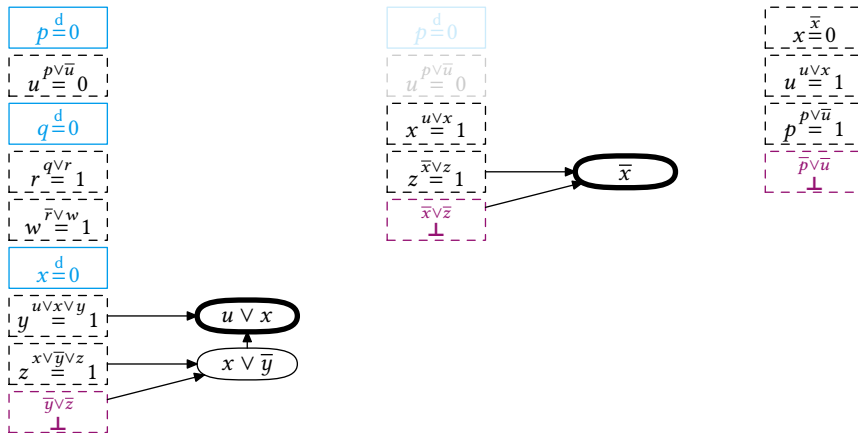
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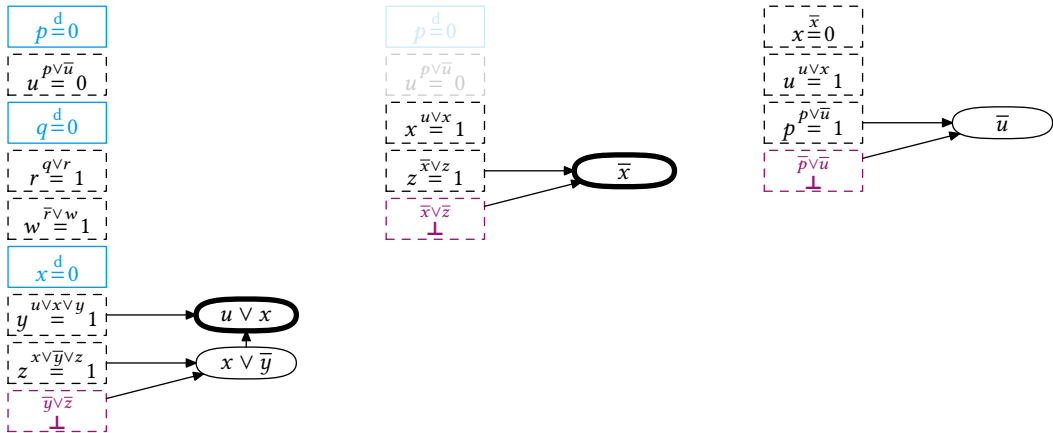
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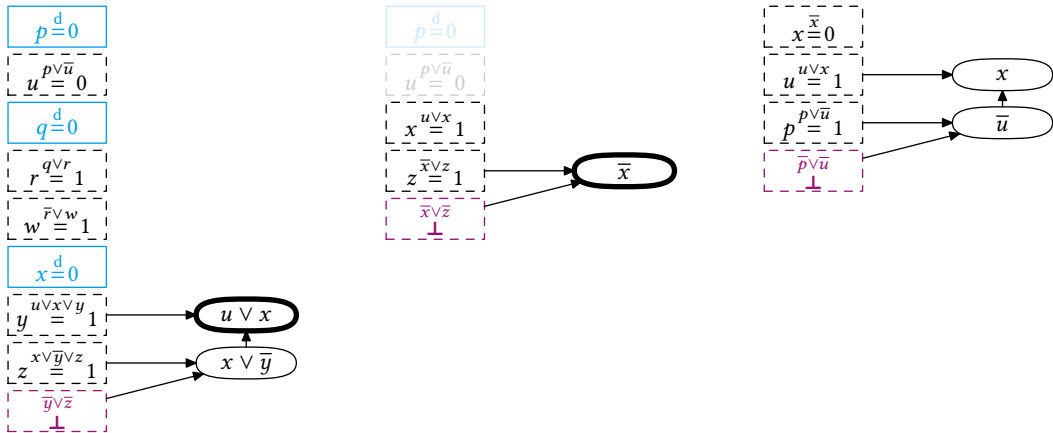
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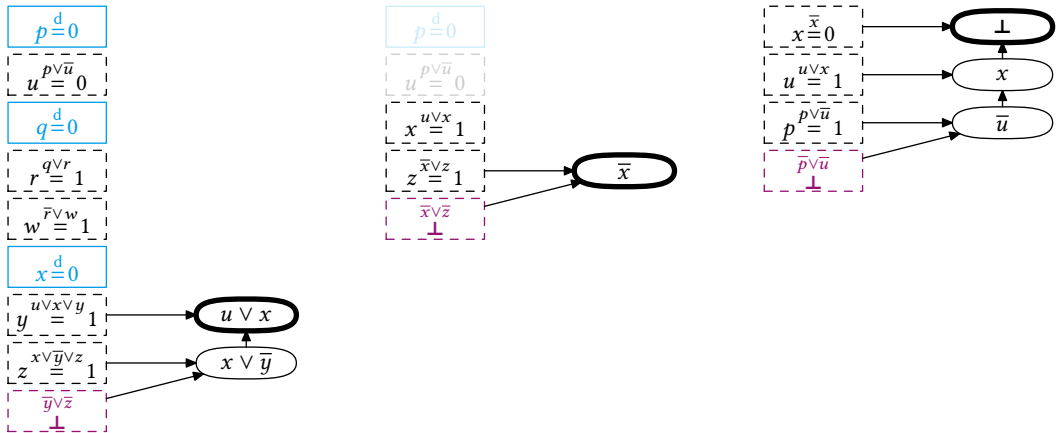
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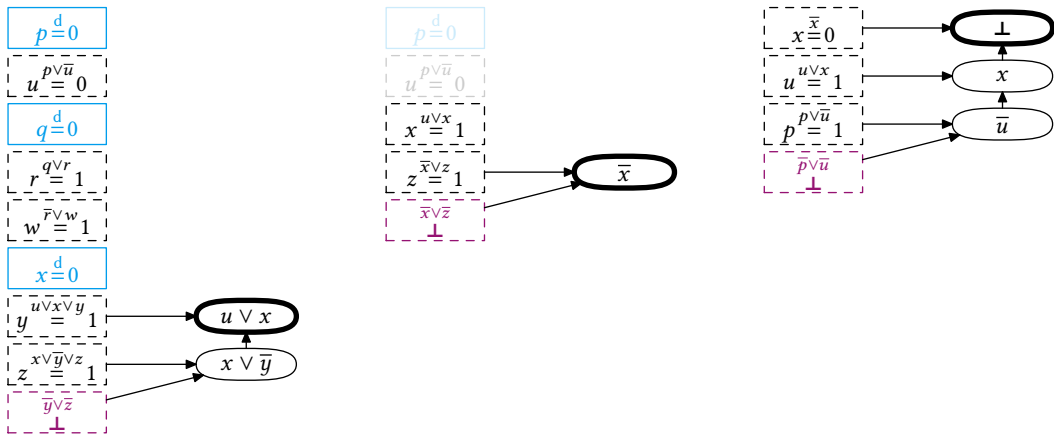
(*) Ignores pre- and inprocessing, but we will get there...

Resolution Proofs from CDCL Executions

Obtain resolution proof...

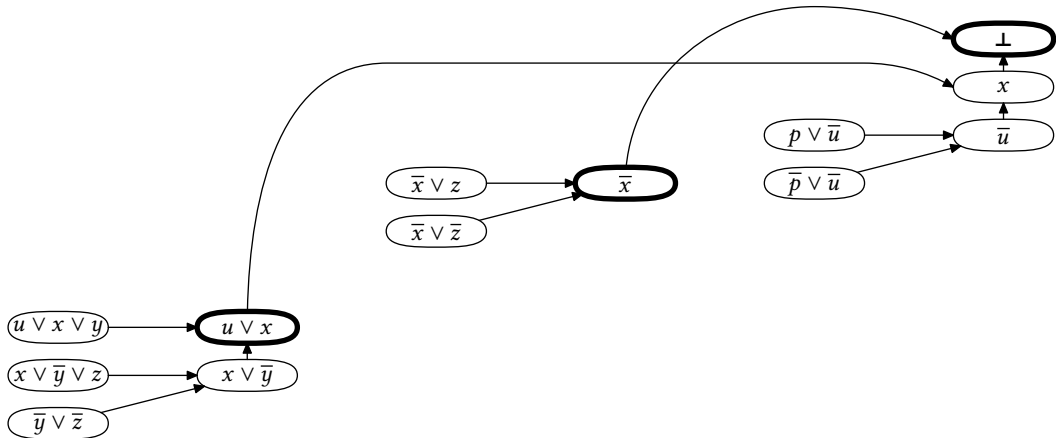
Resolution Proofs from CDCL Executions

Obtain resolution proof from our example CDCL execution...



Resolution Proofs from CDCL Executions

Obtain resolution proof from our example CDCL execution by stringing together conflict analyses:



RUP Proofs and CDCL

But it turns out we can be lazier...

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All learned clauses generated by CDCL solver are RUP clauses

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More Ingredients in Proof Logging for SAT

Fact

RUP proofs can be viewed as shorthand for resolution proofs

See proof complexity and SAT solving survey [BN21] for more on this

But RUP and resolution are not enough for preprocessing, inprocessing, and some other kinds of reasoning

Extension Variables, Part 1

Suppose we want a **reified variable** r encoding

$$r \Leftrightarrow (x \wedge y)$$

Extended resolution [Tse68]

Resolution rule plus **extension rule** introducing clauses

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for fresh variable r (this is fine since r doesn't appear anywhere previously)

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Fact

Extended resolution (RUP + definition of new variables) is essentially equivalent to the DRAT proof logging system most commonly used for SAT solving

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Practical limitations of current SAT proof logging technology:

- Difficulties dealing with stronger reasoning efficiently (even for SAT solving)
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- And more...

Pseudo-Boolean Constraints

0–1 integer linear inequalities or (linear) pseudo-Boolean constraints:

$$\sum_i a_i \ell_i \geq A$$

- $a_i, A \in \mathbb{Z}$
- **literals** ℓ_i : x_i or $\bar{x}_i$ (where $x_i + \bar{x}_i = 1$)

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Sometimes convenient to use **normalised form** [Bar95] with **all a_i, A positive** (without loss of generality)

Some Types of Pseudo-Boolean Constraints

1 Clauses

$$x_1 \vee \bar{x}_2 \vee x_3 \quad \Leftrightarrow \quad x_1 + \bar{x}_2 + x_3 \geq 1$$

2 Cardinality constraints

$$x_1 + x_2 + x_3 + x_4 \geq 2$$

3 General pseudo-Boolean constraints

$$x_1 + 2\bar{x}_2 + 3x_3 + 4\bar{x}_4 + 5x_5 \geq 7$$

Pseudo-Boolean Reasoning: Cutting Planes [CCT87]

Input/model axioms

From the input

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$\ell_i \geq 0$

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$$\frac{\sum_i a_i \ell_i \geq A \quad \sum_i b_i \ell_i \geq B}{\sum_i (a_i + b_i) \ell_i \geq A + B}$$

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Saturation for any $c \in \mathbb{N}^+$
(assumes normalised form)

$$\frac{\sum_i a_i \ell_i \geq A}{\sum_i \min(a_i, A) \cdot \ell_i \geq A}$$

Cutting Planes Toy Example

$$w + 2x + y \geq 2$$

Cutting Planes Toy Example

Multiply by 2 $\frac{w + 2x + y \geq 2}{2w + 4x + 2y \geq 4}$

Cutting Planes Toy Example

$$\text{Multiply by 2} \quad \frac{w + 2x + y \geq 2}{2w + 4x + 2y \geq 4} \quad w + 2x + 4y + 2z \geq 5$$

Cutting Planes Toy Example

$$\begin{array}{rcl} & w + 2x + y \geq 2 & \\ \text{Multiply by 2} & \hline 2w + 4x + 2y \geq 4 & & w + 2x + 4y + 2z \geq 5 \\ \text{Add} & \hline 3w + 6x + 6y + 2z \geq 9 & \end{array}$$

Cutting Planes Toy Example

$$\begin{array}{lcl} \text{Multiply by 2} & \frac{w + 2x + y \geq 2}{2w + 4x + 2y \geq 4} & \\ \text{Add} & \frac{w + 2x + 4y + 2z \geq 5}{3w + 6x + 6y + 2z \geq 9} & \bar{z} \geq 0 \end{array}$$

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 & & \frac{\bar{z} \geq 0}{2\bar{z} \geq 0} \text{ Multiply by 2}
 \end{array}$$

Cutting Planes Toy Example

$$\begin{array}{rcl}
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Add

$$\frac{3w + 6x + 6y}{3w + 6x + 6y} \geq 7$$

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Naming constraints by integers and literal axioms by the literal involved (with ~ for negation) as

Constraint 1

$\doteq 2x + y + w \geq 2$

Constraint 2

$\doteq 2x + 4y + 2z + w \geq 5$

~z

$\doteq \bar{z} \geq 0$

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- Constraint 2 $\doteq 2x + 4y + 2z + w \geq 5$
- $\sim z \doteq \bar{z} \geq 0$

such a calculation is written in the proof log in reverse Polish notation as*

```
pol 1 2 * 2 + ~z 2 * + 3 d ;
```

Cutting Planes Toy Example

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pol

1

2

*

2

+

~z

2

*

+

3

d

;

(*) Except VERIPB variables have to be at least two characters long, but we abuse syntax slightly here for simplicity

Resolution and Cutting Planes

To simulate resolution step such as

$$\frac{\bar{y} \vee \bar{z} \quad x \vee \bar{y} \vee z}{x \vee \bar{y}}$$

we can perform the cutting planes steps

$$\begin{array}{l} \text{Add} \quad \frac{\bar{y} + \bar{z} \geq 1 \quad x + \bar{y} + z \geq 1}{x + 2\bar{y} \geq 1} \\ \text{Divide by 2} \quad \frac{x + 2\bar{y} \geq 1}{x + \bar{y} \geq 1} \end{array}$$

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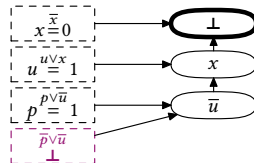
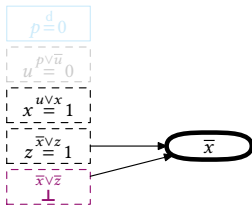
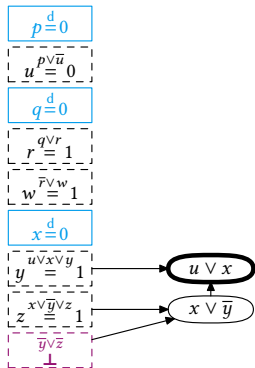
Given that the premises are clauses 7 and 5 in our example CNF formula, using references

$$\begin{array}{l} \text{Constraint 7} \doteq \overline{y} + \overline{z} \geq 1 \\ \text{Constraint 5} \doteq x + \overline{y} + z \geq 1 \end{array}$$

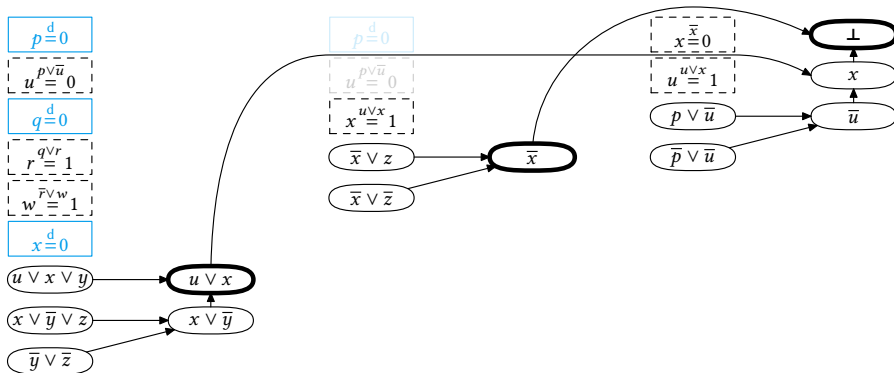
we can write this in the proof log as

$$\text{pol } 7 \ 5 \ + \ 2 \ d \ ;$$

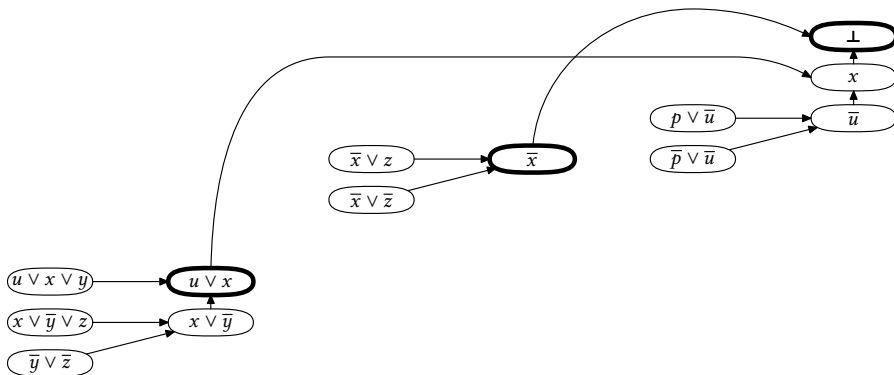
Pseudo-Boolean Proof Logging for Example CDCL Conflict Analyses



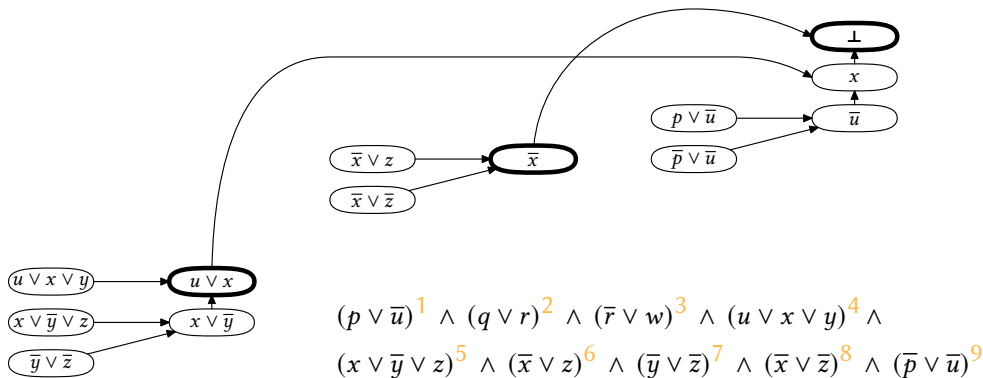
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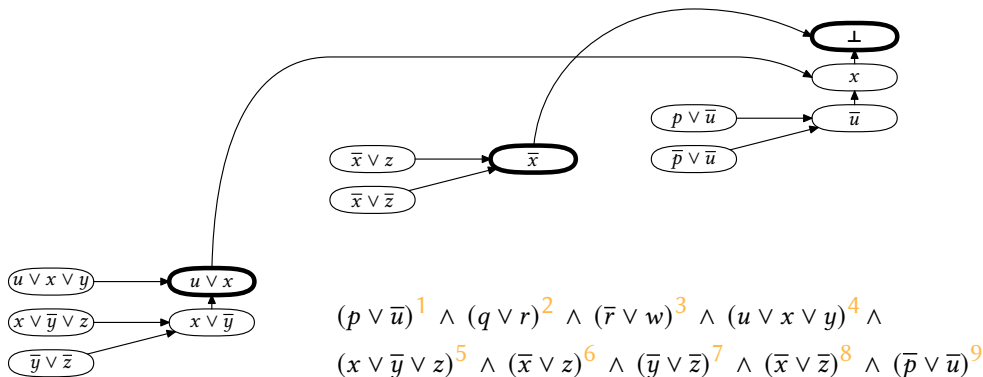
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Pseudo-Boolean Proof Logging for Example CDCL Conflict Analyses



pol 7 5 + 2 d 4 + 2 d ;

pol 8 6 + 2 d ;

pol 9 1 + 2 d 10 + 2 d 11 + 2 d ;

$\rightsquigarrow$ Constraint 10 $\doteq u + x \geq 1$

$\rightsquigarrow$ Constraint 11 $\doteq \bar{x} \geq 1$

$\rightsquigarrow$ Constraint 12 $\doteq 0 \geq 1$ ⚡

RUP Revisited

Can define (reverse) unit propagation in a pseudo-Boolean setting [EGMN20]

Constraint C propagates variable x if setting x to “wrong value” would make C unsatisfiable

E.g., if x_5 is false,

$$x_1 + 2\bar{x}_2 + 3x_3 + 4\bar{x}_4 + 5x_5 \geq 7$$

propagates $\bar{x}_4$ (since other coefficients do not add up to 7)

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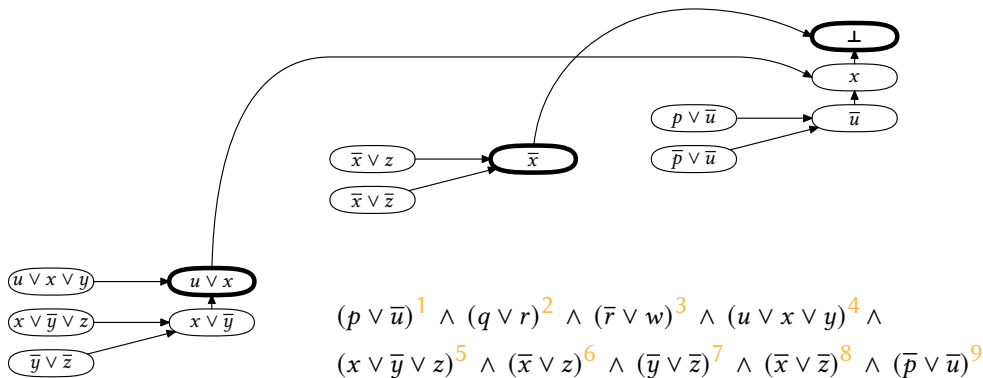
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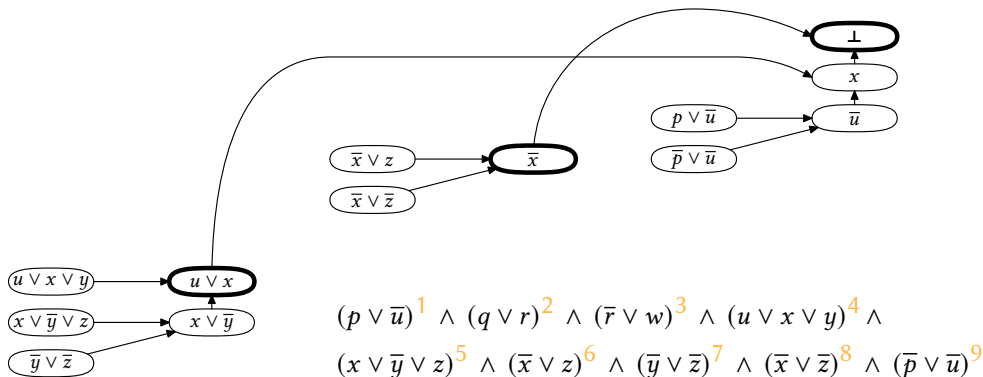
Risk for confusion!

- Constraint programming people might call this (reverse) integer bounds consistency
 - Does the same thing if we're working with clauses
 - More interesting for general pseudo-Boolean constraints
- SAT people beware: constraints can propagate multiple times and multiple variables

Pseudo-Boolean Proof Logging for Example CDCL Execution with RUP



Pseudo-Boolean Proof Logging for Example CDCL Execution with RUP



rup 1 u 1 x >= 1 ;

rup 1 ~x >= 1 ;

rup >= 1 ;

$\rightsquigarrow$ Constraint 10 $\doteq u + x \geq 1$

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Extension Variables, Part 2

Suppose we want new, fresh **reified variable** r encoding

$$r \Leftrightarrow (3x + 2y + z + w \geq 3)$$

This time, introduce constraints

$$3\bar{r} + 3x + 2y + z + w \geq 3 \qquad 5r + 3\bar{x} + 2\bar{y} + \bar{z} + \bar{w} \geq 5$$

Again, needs support from the proof system in the form of **strengthening rules**

Proof Logs for “Cutting Planes with Strengthening”

For satisfiable instances: just specify a satisfying assignment.

For unsatisfiability: a sequence of **pseudo-Boolean constraints** in (slight extension of) **OPB format** [RM16]

- Each constraint follows “obviously” from what is known so far
- Either implicitly, by RUP...
- Or by an explicit cutting planes derivation...
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(*) Not actually implemented this way — more details in a few slides ...

Optimisation and Enumeration Problems

Optimisation

- Define an objective $f = \sum_i w_i \ell_i$, $w_i \in \mathbb{Z}$, to minimise subject to the constraints in the formula
- To maximise, negate objective
- Log solution α with `sol1` rule $\Rightarrow$ objective function value $UB \sum_i w_i \alpha(\ell_i)$
- Add **objective-improving constraint** $\sum_i w_i \ell_i \leq UB - 1$
- Semantics for proof of optimality: “infeasible to find better solution than best so far”
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Enumeration

- When a solution is found, can log it with `solx` rule
- Introduces a new constraint saying “not this solution”
- So the proof semantics is “infeasible, except for all the solutions I told you about”

Deleting Constraints

- In practice, important to erase constraints to save memory and time during verification
- Unsatisfiability proofs: fairly straightforward to deal with from point of view of proof logging
- Optimisation and enumeration proofs: significantly more delicate
- We will mostly ignore deletions during this tutorial for simplicity and clarity

Pseudo-Boolean Proof Logging — How and Why?

- If problem is (special case of) 0-1 integer linear program
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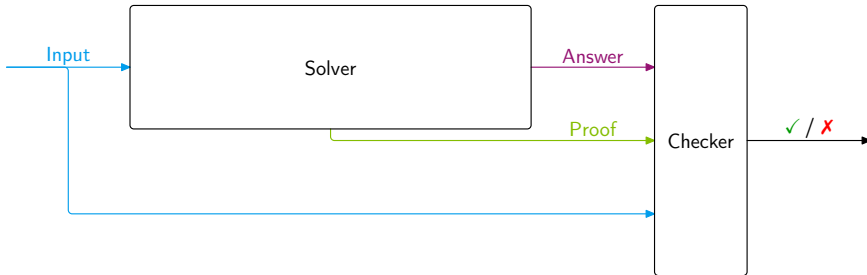
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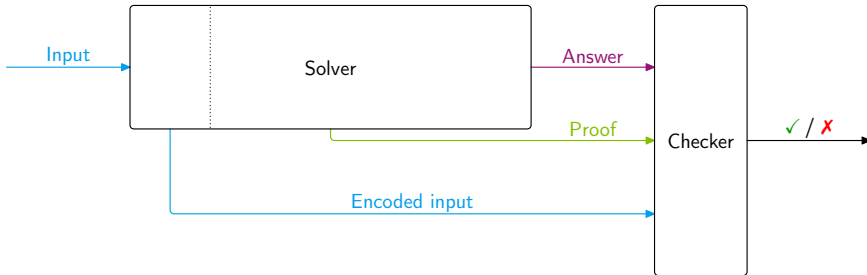
$$7\bar{r} + x_1 + 2\bar{x}_2 + 3x_3 + 4\bar{x}_4 + 5x_5 \geq 7$$

$$9r + \bar{x}_1 + 2x_2 + 3\bar{x}_3 + 4x_4 + 5\bar{x}_5 \geq 9$$

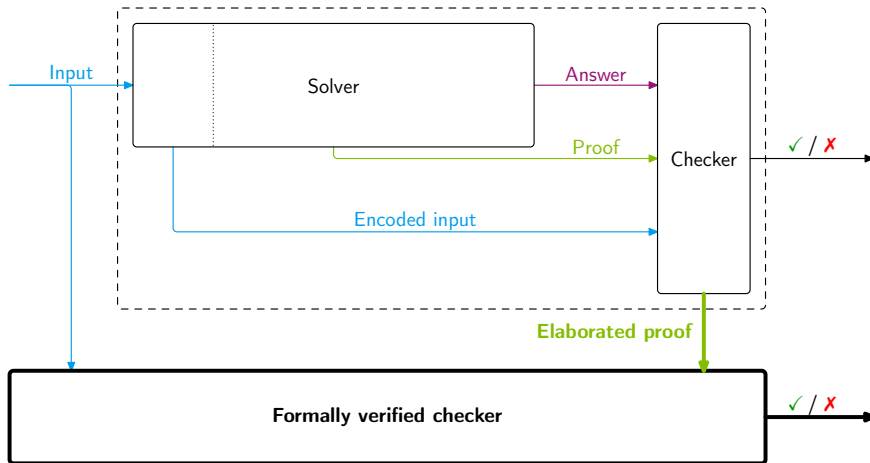
Proof Logging with Formally Verified Checking: Full Workflow



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Wish to allow **without-loss-of-generality** arguments that can derive non-implied constraints

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Deriving $r \Leftrightarrow (3x + 2y + z + w \geq 3)$ Using the Redundance Rule

Want to derive

$$3\bar{r} + 3x + 2y + z + w \geq 3 \quad 5a + 3\bar{x} + 2\bar{y} + \bar{z} + \bar{w} \geq 5$$

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Choose $\omega = \{r \mapsto 0\} - F$ untouched; new constraint satisfied

Deriving $r \Leftrightarrow (3x + 2y + z + w \geq 3)$ Using the Redundance Rule

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VERIPB code:

```
red 3 ~r 3 x 2 y 1 z 1 w >= 3 : r -> 0 ;
red 5 r 3 ~x 2 ~y 1 ~z 1 ~w >= 5 : r -> 1 ;
```

Redundance and Dominance Rules for Optimisation

Redundance-based strengthening, optimisation version

Add constraint C to formula F if exists witness substitution ω s.t.

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- Applying ω should **strictly decrease** f
- If so, don't need to show that $C \upharpoonright_{\omega}$ holds!

Soundness of Dominance Rule

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- 7 ...
- 8 Can't go on forever, so finally reach α' satisfying both F and C

Strength of Dominance Rule

Dominance-based strengthening (stronger, still simplified)

If $C_1, C_2, \dots, C_{m-1}$ have been derived from F (maybe using dominance), then can derive C_m if exists witness substitution ω s.t.

$$F \cup \{C_1, \dots, C_{m-1}\} \cup \{\neg C_m\} \models F \upharpoonright_{\omega} \cup \{f \upharpoonright_{\omega} < f\}$$

Only consider F — no need to show that any $C_i \upharpoonright_{\omega}$ implied!

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Further extensions:

- Define dominance rule w.r.t. order independent of objective
- Switch between different orders in same proof
- See [BGMN23, ABB⁺26b] fore more details

Strengthening Rules: Syntax

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red ⟨Constraint C⟩ : ⟨var1⟩ -> ⟨val1⟩ . . . ⟨varN⟩ -> ⟨valN⟩ : subproof
  subproofs for proof goals
qed;
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- Witness ω should be explicitly specified in proof log
- Subproofs of proof goals should also be explicit
- Except can be skipped for proof goals that “obviously” follow, e.g.,
 - by reverse unit propagation (RUP)
 - by simple syntactic implication from other constraint (as in our reification example)
 - if the proof goal is not affected by the witness substitution ω

Successful Applications of Pseudo-Boolean Proof Logging with VERIPB

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Surprisingly, pseudo-Boolean reasoning with strengthening rules sufficient to efficiently certify wide range of combinatorial solving techniques and paradigms:

- 1 **Boolean satisfiability (SAT) solving** including advanced techniques such as
 - Gaussian elimination [GN21]
 - symmetry breaking [BGMN23, ABB⁺26b]
- 2 **SAT-based optimization (MaxSAT)** [VDB22, BBN⁺23, BBN⁺24, IOT⁺24]
- 3 **Pseudo-Boolean solving and optimization** [GMNO22, KLM⁺25]
- 4 **Graph solving** [GMN20, GMM⁺20, GMM⁺24, DKK⁺26]
- 5 **Dynamic programming and decision diagrams** [DMM⁺24]
- 6 **Presolving in 0–1 integer linear programming** [HOGN24]
- 7 **Constraint programming** [EGMN20, GMN22, MM23, MMN24, MM25]
- 8 **Automated planning** [DHN⁺25]
- 9 **Enumeration of solutions/models** [MNOT26]

VERIPB Resources

VERIPB [tutorials](#), [technical documentation](#), and [more information](#) at
<https://VeriPB.org>



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Lots of [concrete example files](#) at <https://gitlab.com/MIAOresearch/software/VeriPB>

Summing up the First Lecture



- Combinatorial solving and optimisation is a true success story
- But ensuring correctness is a crucial, and not yet satisfactorily addressed, concern
- Certifying solvers producing machine-verifiable proofs of correctness seems like most promising approach
- Cutting planes reasoning with pseudo-Boolean constraints seems to hit a sweet spot between simplicity and expressivity

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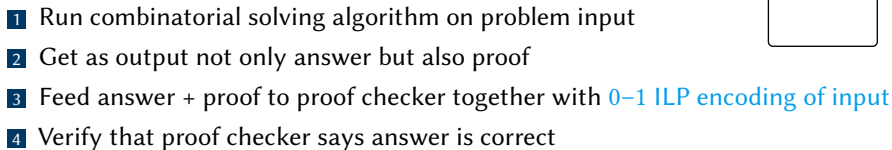
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Questions so far?



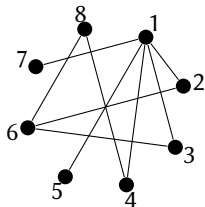
VERIPB Proof Structure: Syntax

```

pseudo-Boolean proof version 3.0
f  $\langle M \rangle$ 
 $\langle derivation\ part \rangle$ 
output  $\langle output\ part \rangle$ 
conclusion  $\langle conclusion\ part \rangle$ 
end pseudo-Boolean proof

```

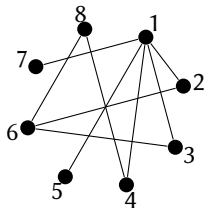

Saturation for any $c \in \mathbb{N}^+$
(assumes normalised form)



...

More Advanced Example of Redundance-Based Strengthening

Is the following graph 2-colourable (bipartite)?



$$b_1 + b_2 \geq 1$$

$$\bar{b}_1 + \bar{b}_2 \geq 1$$

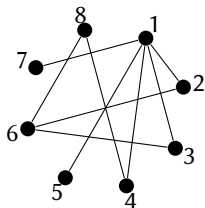
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Without loss of generality, node 1 is blue (otherwise we can swap the two colours)

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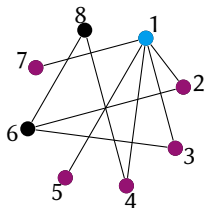
• • •

$$F = F \upharpoonright_\omega \quad \text{and} \quad \neg(b_1 \geq 1) = (b_1 \geq 1) \upharpoonright_\omega$$

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pseudo-Boolean proof version 3.0

f 18

```
red 1 b1 >= 1 : b1 -> ~b1 b2 -> ~b2
```

→ $b_3 \rightarrow \sim b_3 \quad b_4 \rightarrow \sim b_4 \quad b_5 \rightarrow \sim b_5$

→ $b_6 \rightarrow \sim b_6 \quad b_7 \rightarrow \sim b_7 \quad b_8 \rightarrow \sim b_8 ;$

```
rup >= 1 ;
```

```
output NONE
```

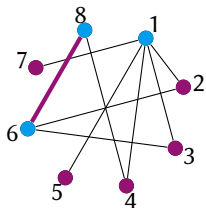
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end pseudo-Boolean proof
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- Start the proof
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- Derive contradiction by RUP

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Derive $b_1 \geq 1$ with redundancy; witness
 $\omega : b_i \mapsto \bar{b}_i$ for all i

$$F = F \upharpoonright_\omega \quad \text{and} \quad \neg(b_1 \geq 1) = (b_1 \geq 1) \upharpoonright_\omega$$

pseudo-Boolean proof version 3.0

f 18

```
red 1 b1 >= 1 : b1 -> ~b1 b2 -> ~b2
```

→ $b_3 \rightarrow \sim b_3 \quad b_4 \rightarrow \sim b_4 \quad b_5 \rightarrow \sim b_5$

→ $b_6 \rightarrow \sim b_6 \quad b_7 \rightarrow \sim b_7 \quad b_8 \rightarrow \sim b_8 ;$

```
rup >= 1 ;
```

```
output NONE
```

conclusion UNSAT

end pseudo-Boolean proof

- Start the proof
- Derive $b_1 \geq 1$ with redundancy rule
- Derive contradiction by RUP

The Problem of Deleting Constraints

Important to allow deletions of constraints from database

- Improves practical performance
- Affects power of strengthening rules

But powerful strengthening rules create problems:

- Unsatisfiable formulas can turn satisfiable
- Satisfiable formulas can turn unsatisfiable(!)
- Optimisation: can create too-good-to-be-true solutions
- Enumeration: can reintroduce already enumerated solutions

Status is undetermined

Propagate given assignment w.r.t. database, then check against original formula

If no assignment given, then

- solution should have been logged
- no unchecked deletion must have occurred

Only valid if no solution has been logged

Check that specified constraint is contradictory (technically: negative slack)

If no constraint specified, check that database contains contradictory constraint



$$\rightsquigarrow 0 \geq 1 \quad \text{⚡}$$

Linear Pseudo-Boolean Constraints and Normalised Form

Usually, **pseudo-Boolean constraints** are any **0-1 integer linear constraints** on the form

$$\sum_i a_i \ell_i \asymp A$$

- $\bowtie \in \{\geq, \leq, =, >, <\}$
- $a_i, A \in \mathbb{Z}$
- **literals** ℓ_i : x_i or $\bar{x}_i$ (where $x_i + \bar{x}_i = 1$)
- variables x_i take values $0 = \text{false}$ or $1 = \text{true}$

Conversion to Normalised Form: Example

Normalised form is used for convenience and without loss of generality

$$-x_1 + 2x_2 - 3x_3 + 4x_4 - 5x_5 < 0$$

- 1** Make inequality non-strict

$$-x_1 + 2x_2 - 3x_3 + 4x_4 - 5x_5 \leq -1$$

- 2** Multiply by -1 to get greater-than-or-equal

$$x_1 - 2x_2 + 3x_3 - 4x_4 + 5x_5 \geq 1$$

Normalised form is used for convenience and without loss of generality

1 Make inequality non-strict

2 Multiply by -1 to get greater-than-or-equal

3 Replace $-\ell$ by $-(1 - \bar{\ell})$ [where we define $\bar{\bar{x}} \doteq x$]

4 Replace “=” by two inequalities “ $\geq$ ” and “ $\leq$ ”

Conversion to Normalised Form: Formal Details

Given (normalised) linear form $\sum_i a_i \ell_i$ with coefficient sum $\sum_i a_i = W$

Syntactic sugar	Meaning
$\sum_i a_i \ell_i > A$	$\sum_i a_i \ell_i \geq A + 1$
$\sum_i a_i \ell_i \leq A$	$\sum_i a_i \bar{\ell}_i \geq W - A$
$\sum_i a_i \ell_i < A$	$\sum_i a_i \bar{\ell}_i \geq W - A + 1$
$\sum_i a_i \ell_i = A$	$\sum_i a_i \ell_i \geq A \quad \text{and} \quad \sum_i a_i \bar{\ell}_i \geq W - A$

Conversion to Normalised Form: Formal Details

Given (normalised) linear form $\sum_i a_i \ell_i$ with coefficient sum $\sum_i a_i = W$

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$\sum_i a_i \ell_i < A$	$\sum_i a_i \bar{\ell}_i \geq W - A + 1$
$\sum_i a_i \ell_i = A$	$\sum_i a_i \ell_i \geq A \quad \text{and} \quad \sum_i a_i \bar{\ell}_i \geq W - A$

In what follows:

- Use syntactic sugar when convenient
- Assume (implicit) normalisation whenever it matters
- Write $\doteq$ for syntactic equality

Still with $C \doteq x_1 + 2x_2 + 3x_3 + 4x_4 \geq 5$, so that $\neg C \doteq \bar{x}_1 + 2\bar{x}_2 + 3\bar{x}_3 + 4\bar{x}_4 \geq 6$
(and recall that $\ell + \bar{\ell} = 1$, so that opposing literals cancel and leave 1)

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Derivations Under Assumptions: The General Claim

This works in general!

- Suppose that we have
 - 1 constraints $C_1, C_2, \dots, C_m$
 - 2 assumptions $\rho = \{\ell_1 = 1, \ell_2 = 1, \dots, \ell_n = 1\}$
 - 3 a derivation of D from the restricted constraints $C_1 \upharpoonright_\rho, C_2 \upharpoonright_\rho, \dots, C_m \upharpoonright_\rho$

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- Then the same derivation applied to $C_1, C_2, \dots, C_m$ yields a constraint at least as strong as

$$\ell_1 \wedge \ell_2 \wedge \cdots \wedge \ell_n \Rightarrow D \doteq \deg(D) \cdot \sum_{i=1}^n \bar{\ell}_i + D$$

Derivations Under Assumptions: The General Claim

This works in general!

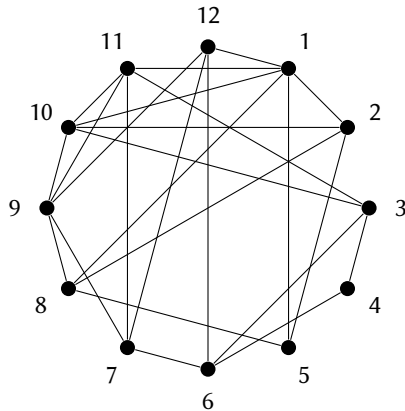
- Suppose that we have
 - 1 constraints $C_1, C_2, \dots, C_m$
 - 2 assumptions $\rho = \{\ell_1 = 1, \ell_2 = 1, \dots, \ell_n = 1\}$
 - 3 a derivation of D from the restricted constraints $C_1 \upharpoonright_\rho, C_2 \upharpoonright_\rho, \dots, C_m \upharpoonright_\rho$
- Then the same derivation applied to $C_1, C_2, \dots, C_m$ yields a constraint at least as strong as

$$\ell_1 \wedge \ell_2 \wedge \cdots \wedge \ell_n \Rightarrow D \doteq \deg(D) \cdot \sum_{i=1}^n \bar{\ell}_i + D$$

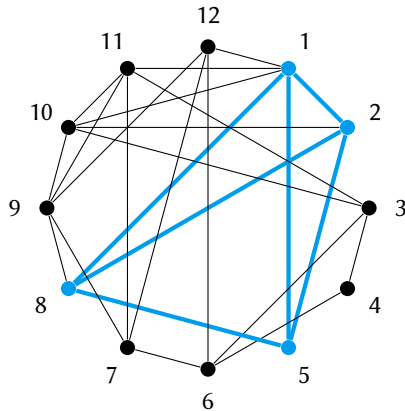
Give a convenient way to construct derivations:

- Apply assumptions in ρ to constraints C_j and simplify to get $C_j \upharpoonright_\rho$
- Derive desired conclusion D
- Remove assumptions to get reified version $\rho \Rightarrow D$ of constraints

The Maximum Clique Problem

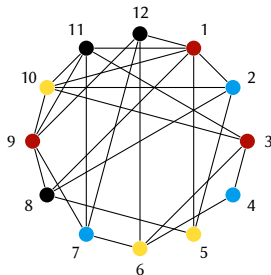


The Maximum Clique Problem



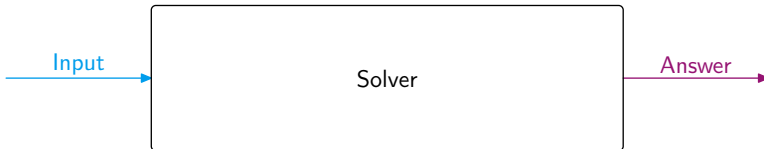
- Pick a vertex v
- Either v is in the clique...
 - Throw away every vertex not adjacent to v
 - If vertices remain, recurse
- ...or v is not in the clique
 - Throw away v (and all incident edges) and pick another vertex

A Brief and Incomplete Guide to Clique Solving (3/4)

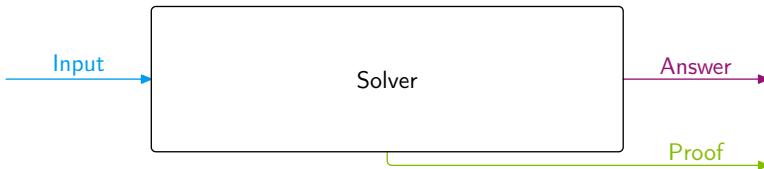


Given a k -colouring of a subgraph, that subgraph cannot have a clique of more than k vertices

We can use $|C| + \#colours(P)$ as a bound, for any colouring

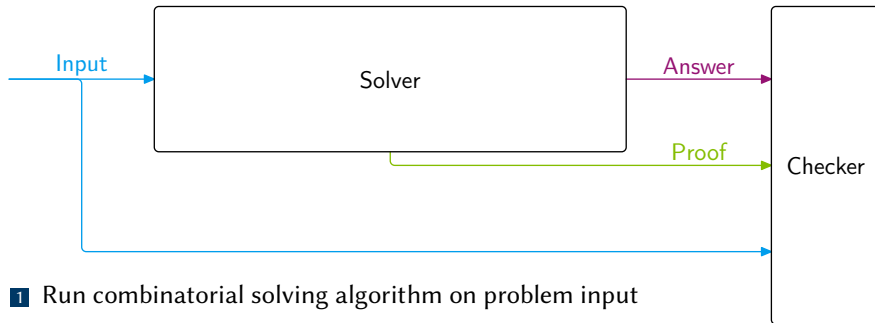


- 1 Run combinatorial solving algorithm on problem input



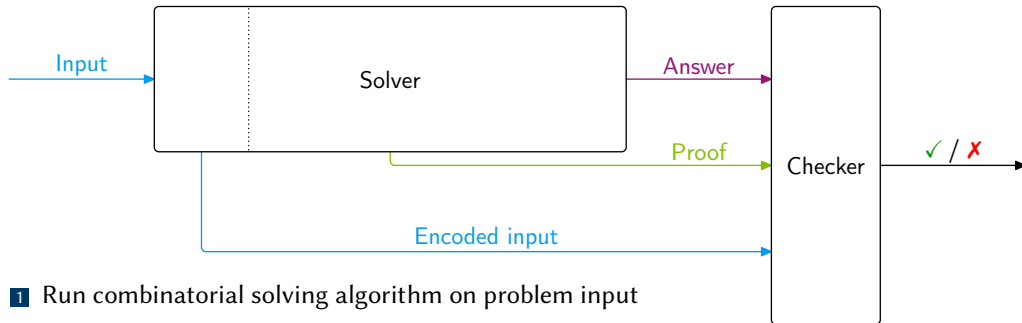
- 1 Run combinatorial solving algorithm on problem input
- 2 Get as output not only answer but also proof

A Slightly Different Proof Logging Workflow



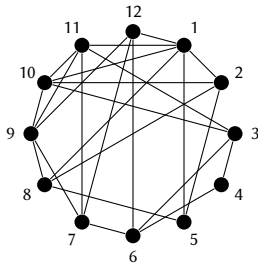
- 1 Run combinatorial solving algorithm on problem input
- 2 Get as output not only answer but also proof
- 3 Feed answer + proof to proof checker together with input

A Slightly Different Proof Logging Workflow



- 1 Run combinatorial solving algorithm on problem input
- 2 Get as output not only answer but also proof
- 3 Feed answer + proof to proof checker together with 0-1 ILP encoding of input
- 4 Verify that proof checker says answer is correct

A Pseudo-Boolean Encoding for Clique (in OPB Format)



```
* #variable= 12 #constraint= 41
```

```
min: 1 ~x1 1 ~x2 1 ~x3 1 ~x4 . . . and so on. . . 1 ~x11 1 ~x12 ;
```

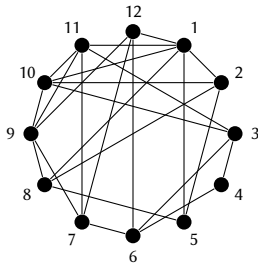
```
@noedge1_3 -1 x3 -1 x1 >= -1 ;
```

```
@noedge2_3 -1 x3 -1 x2 >= -1 ;
```

```
@noedge1_4 -1 x4 -1 x1 >= -1 ;
```

```
* . . . and a further 38 similar lines for the remaining non-edges
```

A Pseudo-Boolean Encoding for Clique (in OPB Format)



* #variable= 12 #constraint= 41

min: 1 ~x1 1 ~x2 1 ~x3 1 ~x4 . . . and so on. . . 1 ~x11 1 ~x12 ;

@noedge1_3 -1 x3 -1 x1 >= -1 ;

@noedge2_3 -1 x3 -1 x2 >= -1 ;

@noedge1_4 -1 x4 -1 x1 >= -1 ;

* . . . and a further 38 similar lines for the remaining non-edges

$x_i = 1$ means “vertex i is in the clique”

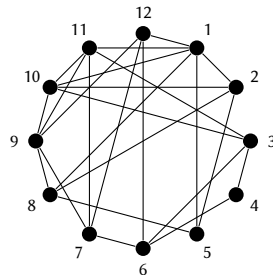
VERIPB proof minimises number of vertices **not** selected, so 4-clique yields objective value $12 - 4 = 8$

First Attempt at a Proof

```

pseudo-Boolean proof version 3.0
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;

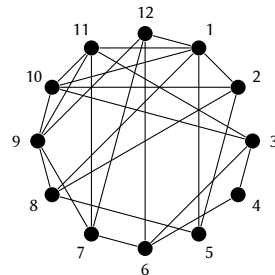
```



First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```



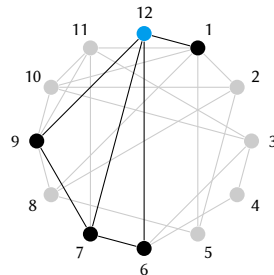
Start with a header

The problem constraints are loaded for us

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```

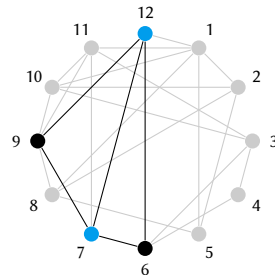


Branch accepting 12
Throw away non-adjacent vertices

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```

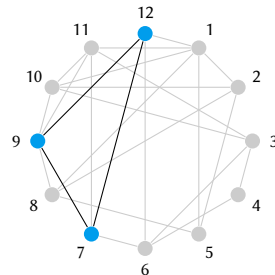


Branch also accepting 7
Throw away non-adjacent vertices

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```

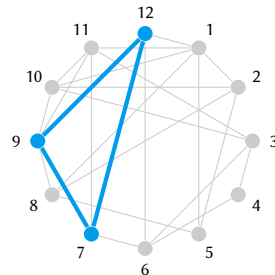


Branch also accepting 9
Throw away non-adjacent vertices

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```



We branched on 12, 7, 9
 Found a new incumbent
 Now looking for a ≥ 4 vertex clique
 “@obj” names the constraint this creates

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
```

```
rup 1 ~x12 1 ~x7 >= 1 ;
```

```
rup 1 ~x12 >= 1 ;
```

```
rup 1 ~x11 1 ~x10 >= 1 ;
```

```
rup 1 ~x11 >= 1 ;
```

```
@obj soli x1 x2 x5 x8 ;
```

```
rup 1 ~x8 1 ~x5 >= 1 ;
```

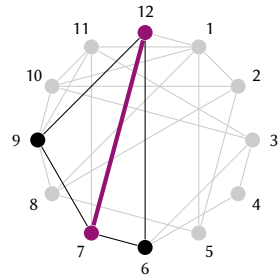
```
rup 1 ~x8 >= 1 ;
```

```
rup >= 1 ;
```

```
output NONE ;
```

```
conclusion BOUNDS 8 8 ;
```

```
end pseudo-Boolean proof ;
```

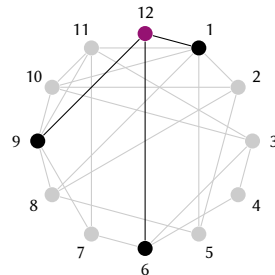


Backtrack from 12, 7
 9 explored already, only 6 feasible
 No ≥ 4 vertex clique possible
 Effectively this deletes the 7–12 edge

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```

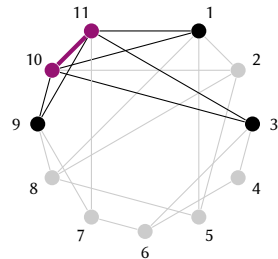


Backtrack from 12
 Only 1, 6 and 9 feasible (1-colourable)
 No ≥ 4 vertex clique possible
 Effectively this deletes vertex 12

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```

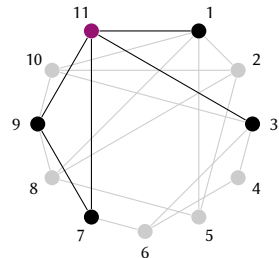


Branch on 11 then 10
 Only 1, 3 and 9 feasible (1-colourable)
 No ≥ 4 vertex clique possible
 Backtrack, deleting the edge

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```

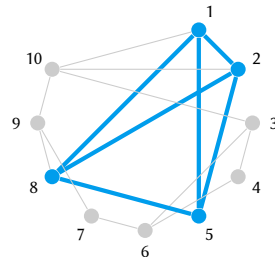


Backtrack from 11
2-colourable, so no ≥ 4 clique
Delete the vertex

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```

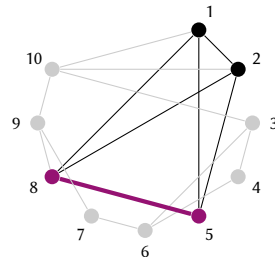


Branch on 8, 5, 1, 2
Find a new incumbent
Now looking for a ≥ 5 vertex clique

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```

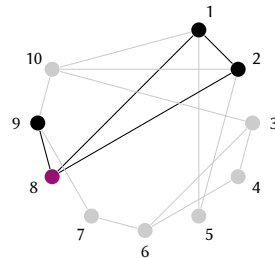


Backtrack from 8, 5
Only 4 vertices; can't have a ≥ 5 clique
Delete the edge

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```

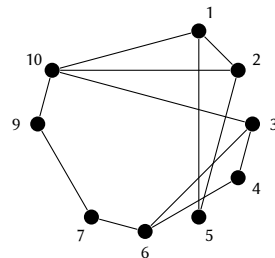


Backtrack from 8
Still not enough vertices
Delete the vertex

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```



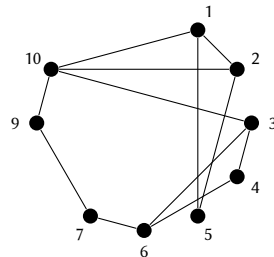
Remaining graph is 3-colourable
Backtrack from root node

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
```

```
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```



Finish with what we've concluded
 We specify a lower and an upper bound
 Remember we're minimising the number of vertices we *don't* select, so a 4-clique has an objective value of $12 - 4 = 8$
 ("output" is for features we're not using)

Verifying This Proof (Or Not...)

```
$ veripb clique.opb clique-attempt-one.pbp
Running VeriPB version 3.0.2
Error: Checking error at clique-attempt-one.pbp:5
```

Caused by:

```
The constraint is not implied by reverse unit propagation
(RUP) from core and derived database. Rerun with the option
'--trace-failed' to see the propagations. . . .
```

Verifying This Proof (Or Not...)

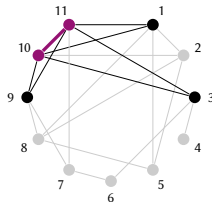
```
$ veripb clique.opb clique-attempt-one.pbp
```

```
Running VeriPB version 3.0.2
```

```
Error: Checking error at clique-attempt-one.pbp:5
```

Caused by:

The constraint is not implied by reverse unit propagation (RUP) from core and derived database. Rerun with the option '--trace-failed' to see the propagations. . . .



Verifying This Proof (Or Not...)

```
$ veripb --trace clique.opb clique-attempt-one.pbp
Running VeriPB version 3.0.2
  Objective: min 1 ~x1 1 ~x2 . . . 1 ~x11 1 ~x12 0
  ConstraintId 1: 1 ~x1 1 ~x3 >= 1
  ConstraintId 2: 1 ~x2 1 ~x3 >= 1
  . . .
  ConstraintId 41: 1 ~x11 1 ~x12 >= 1
line   1: pseudo-Boolean proof version 3.0
line   2: @obj soli x7 x9 x12 ;
  ConstraintId 42: 1 x1 1 x2 . . . 1 x11 1 x12 >= 4
line   3: rup 1 ~x12 1 ~x7 >= 1 ;
  ConstraintId 43: 1 ~x7 1 ~x12 >= 1
line   4: rup 1 ~x12 >= 1 ;
  ConstraintId 44: 1 ~x12 >= 1
line   5: rup 1 ~x11 1 ~x10 >= 1 ;
Error: Checking error at clique-attempt-one.pbp:5
```

Verifying This Proof (Or Not...)

```
$ veripb --trace-failed clique.opb clique-attempt-one.pbp
Running VeriPB version 3.0.2
Propagation check failed! The propagation had the following trail:
  propagations in format: <assignment> (<reason constraint>)
    ~x12 (1 ~x12 >= 1)
    x10 (1 x10 1 x11 >= 2)
    x11 (1 x10 1 x11 >= 2)
    ~x4 (1 ~x4 1 ~x10 >= 1)
    ~x5 (1 ~x5 1 ~x10 >= 1)
    ~x6 (1 ~x6 1 ~x10 >= 1)
    ~x7 (1 ~x7 1 ~x10 >= 1)
    ~x8 (1 ~x8 1 ~x10 >= 1)
    ~x2 (1 ~x2 1 ~x11 >= 1)
Error: Checking error at clique-attempt-one.pbp:5
```

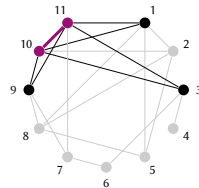
The Problem of Counting

The proof checker isn't smart enough to work out that

- vertices 1, 3 and 9 can all be given the same colour...
- and each colour class contributes at most one vertex to a clique...
- so “find me a clique with more than three vertices” can't be satisfied once we've accepted both 10 and 11

Unit propagation never sees this:

- The bound needs **counting**
- But the encoding only ever talks about **pairs** of vertices



Dealing With Colourings

So the colour bound doesn't follow by reverse unit propagation...

But we can lazily recover at-most-one constraints for each colour class!

$$\begin{aligned}
 &(\bar{x}_1 + \bar{x}_6 \geq 1) \\
 + &(\bar{x}_1 + \bar{x}_9 \geq 1) &&= 2\bar{x}_1 + \bar{x}_6 + \bar{x}_9 \geq 2 \\
 + &(\bar{x}_6 + \bar{x}_9 \geq 1) &&= 2\bar{x}_1 + 2\bar{x}_6 + 2\bar{x}_9 \geq 3 \\
 &&&/ 2 &&= \bar{x}_1 + \bar{x}_6 + \bar{x}_9 \geq 2 \\
 &&&&&\text{i.e. } x_1 + x_6 + x_9 \leq 1
 \end{aligned}$$

Dealing With Colourings

So the colour bound doesn't follow by reverse unit propagation...

But we can lazily recover at-most-one constraints for each colour class!

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 &(\bar{x}_1 + \bar{x}_6 \geq 1) \\
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 + &(\bar{x}_6 + \bar{x}_9 \geq 1) &&= 2\bar{x}_1 + 2\bar{x}_6 + 2\bar{x}_9 \geq 3 \\
 &&&/ 2 &&= \bar{x}_1 + \bar{x}_6 + \bar{x}_9 \geq 2 \\
 &&&&&\text{i.e. } x_1 + x_6 + x_9 \leq 1
 \end{aligned}$$

This generalises to colour classes of any size v

- Each non-edge is used exactly once, $v(v-1)$ additions
- $v-3$ multiplications and $v-2$ divisions

Solvers don't need to "understand" cutting planes to write out this derivation

Details on At-Most-One Constraint Recovery (3 Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
- Need this to prove that **at most one vertex per colour class** can be a member of the clique

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- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\text{Add} \quad \frac{\bar{x}_1 + \bar{x}_2 \geq 1 \quad \bar{x}_1 + \bar{x}_3 \geq 1}{2\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2}$$

Details on At-Most-One Constraint Recovery (3 Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
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- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\begin{array}{rcl}
 & \bar{x}_1 + \bar{x}_2 \geq 1 & \bar{x}_1 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2 & \bar{x}_2 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 3
 \end{array}$$

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- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\begin{array}{rcl}
 & \bar{x}_1 + \bar{x}_2 \geq 1 & \bar{x}_1 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2 & \bar{x}_2 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 3 \\
 \text{Divide by 2} & \hline
 & \bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq \frac{3}{2}
 \end{array}$$

Details on At-Most-One Constraint Recovery (3 Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\begin{array}{rcl}
 & \bar{x}_1 + \bar{x}_2 \geq 1 & \bar{x}_1 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2 & \bar{x}_2 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 3 \\
 \text{Divide by 2} & \hline
 & \bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq \left\lceil \frac{3}{2} \right\rceil
 \end{array}$$

Details on At-Most-One Constraint Recovery (3 Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\begin{array}{rcl}
 & \bar{x}_1 + \bar{x}_2 \geq 1 & \bar{x}_1 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2 & \bar{x}_2 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 3 \\
 \text{Divide by 2} & \hline
 & \bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2
 \end{array}$$

Details on At-Most-One Constraint Recovery (3 Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\begin{array}{rcl}
 & \bar{x}_1 + \bar{x}_2 \geq 1 & \bar{x}_1 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2 & \bar{x}_2 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 3 \\
 \text{Divide by 2} & \hline
 & \bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2
 \end{array}$$

Assuming constraint $\bar{x}_i + \bar{x}_j \geq 1$ has label @C*ij*, get VERiPB code

```
pol @C12 @C13 + @C23 + 2 d ;
```


Details on At-Most-One Constraint Recovery (4 and More Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2$$

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- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\text{Multiply by 2} \quad \frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2}{2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4}$$

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- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\begin{array}{rcl}
 \text{Multiply by 2} & \frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2}{2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4} & \bar{x}_1 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4}{3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5} & \bar{x}_2 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5}{3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6} & \bar{x}_3 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6}{3\bar{x}_1 + 3\bar{x}_2 + 3\bar{x}_3 + 3\bar{x}_4 \geq 7} &
 \end{array}$$

Details on At-Most-One Constraint Recovery (4 and More Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\begin{array}{rcl}
 & \bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2 & \\
 \text{Multiply by 2} & \hline
 2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4 & \bar{x}_1 + \bar{x}_4 \geq 1 & \\
 \text{Add} & \hline
 3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5 & \bar{x}_2 + \bar{x}_4 \geq 1 & \\
 \text{Add} & \hline
 3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6 & \bar{x}_3 + \bar{x}_4 \geq 1 & \\
 \text{Add} & \hline
 3\bar{x}_1 + 3\bar{x}_2 + 3\bar{x}_3 + 3\bar{x}_4 \geq 7 & & \\
 \text{Divide by 3} & \hline
 \bar{x}_1 + \bar{x}_2 + \bar{x}_3 + \bar{x}_4 \geq \left\lceil \frac{7}{3} \right\rceil & &
 \end{array}$$

Details on At-Most-One Constraint Recovery (4 and More Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\begin{array}{rcl}
 & \bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2 & \\
 \text{Multiply by 2} & \hline
 2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4 & \bar{x}_1 + \bar{x}_4 \geq 1 & \\
 \text{Add} & \hline
 3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5 & \bar{x}_2 + \bar{x}_4 \geq 1 & \\
 \text{Add} & \hline
 3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6 & \bar{x}_3 + \bar{x}_4 \geq 1 & \\
 \text{Add} & \hline
 3\bar{x}_1 + 3\bar{x}_2 + 3\bar{x}_3 + 3\bar{x}_4 \geq 7 & & \\
 \text{Divide by 3} & \hline
 \bar{x}_1 + \bar{x}_2 + \bar{x}_3 + \bar{x}_4 \geq 3 & &
 \end{array}$$

Details on At-Most-One Constraint Recovery (4 and More Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
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- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\begin{array}{rcl}
 \text{Multiply by 2} & \frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2}{2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4} & \bar{x}_1 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4}{3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5} & \bar{x}_2 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5}{3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6} & \bar{x}_3 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6}{3\bar{x}_1 + 3\bar{x}_2 + 3\bar{x}_3 + 3\bar{x}_4 \geq 7} & \\
 \text{Divide by 3} & \frac{3\bar{x}_1 + 3\bar{x}_2 + 3\bar{x}_3 + 3\bar{x}_4 \geq 7}{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 + \bar{x}_4 \geq 3} &
 \end{array}$$

Assuming constraint $\bar{x}_i + \bar{x}_j \geq 1$ has label @C_{ij}, VERIPB code for full derivation is

```
pol @C12 @C13 + @C23 + 2 d 2 * @C14 + @C24 + @C34 + 3 d ;
```

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- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
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$$\begin{array}{rcl}
 \text{Multiply by 2} & \frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2}{2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4} & \bar{x}_1 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4}{3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5} & \bar{x}_2 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5}{3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6} & \bar{x}_3 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6}{3\bar{x}_1 + 3\bar{x}_2 + 3\bar{x}_3 + 3\bar{x}_4 \geq 7} & \\
 \text{Divide by 3} & \frac{3\bar{x}_1 + 3\bar{x}_2 + 3\bar{x}_3 + 3\bar{x}_4 \geq 7}{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 + \bar{x}_4 \geq 3} &
 \end{array}$$

Assuming constraint $\bar{x}_i + \bar{x}_j \geq 1$ has label @C ij , VERIPB code for full derivation is

```
pol @C12 @C13 + @C23 + 2 d 2 * @C14 + @C24 + @C34 + 3 d ;
```

(and this pattern generalizes to any number of variables)

What The Revised Proof with Explicit Counting Looks Like

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
pol @noedge1_6 @noedge1_9 + @noedge6_9 + 2 d @obj + ;
rup 1 ~x12 >= 1 ;
pol @noedge1_3 @noedge1_9 + @noedge3_9 + 2 d @obj + ;
rup 1 ~x11 1 ~x10 >= 1 ;
pol @noedge1_3 @noedge1_7 + @noedge3_7 + 2 d @obj + ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
pol @obj @noedge1_9 + ;
rup 1 ~x8 >= 1 ;
pol @noedge1_3 @noedge1_7 + @noedge3_7 + 2 d @noedge2_4 @noedge2_9 + @noedge4_9 + 2 d +
    @noedge5_6 @noedge5_10 + @noedge6_10 + 2 d + @obj + ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```


What The Revised Proof with Explicit Counting Looks Like

pseudo-Boolean proof version 3.0

@obj soli x7 x9 x12 ;

rup 1 ~x12 1 ~x7 >= 1 ;

pol @noedge1_6 @noedge1_9 + @noedge6_9 + 2 d @obj + ;

rup 1 ~x12 >= 1 ;

pol @noedge1_3 @noedge1_9 + @noedge3_9 + 2 d @obj + ;

rup 1 ~x11 1 ~x10 >= 1 ;

pol @noedge1_3 @noedge1_7 + @noedge3_7 + 2 d @obj + ;

rup 1 ~x11 >= 1 ;

@obj soli x1 x2 x5 x8 ;

rup 1 ~x8 1 ~x5 >= 1 ;

pol @obj @noedge1_9 + ;

rup 1 ~x8 >= 1 ;

pol @noedge1_3 @noedge1_7 + @noedge3_7 + 2 d @noedge2_4 @noedge2_9 + @noedge4_9 + 2 d +

@noedge5_6 @noedge5_10 + @noedge6_10 + 2 d + @obj + ;

rup >= 1 ;

output NONE ;

conclusion BOUNDS 8 8 ;

end pseudo-Boolean proof ;

Failing RUP steps have become **pol** derivations: add three non-adjacency constraints together, divide by two, then add the result to the solution-improving constraint

The labels come straight out of the OPB file

What The Revised Proof with Explicit Counting Looks Like

pseudo-Boolean proof version 3.0

@obj soli x7 x9 x12 ;

rup 1 ~x12 1 ~x7 >= 1 ;

pol @noedge1_6 @noedge1_9 + @noedge6_9 + 2 d @obj + ;

rup 1 ~x12 >= 1 ;

pol @noedge1_3 @noedge1_9 + @noedge3_9 + 2 d @obj + ;

rup 1 ~x11 1 ~x10 >= 1 ;

pol @noedge1_3 @noedge1_7 + @noedge3_7 + 2 d @obj + ;

rup 1 ~x11 >= 1 ;

@obj soli x1 x2 x5 x8 ;

rup 1 ~x8 1 ~x5 >= 1 ;

pol @obj @noedge1_9 + ;

rup 1 ~x8 >= 1 ;

pol @noedge1_3 @noedge1_7 + @noedge3_7 + 2 d @noedge2_4 @noedge2_9 + @noedge4_9 + 2 d +
@noedge5_6 @noedge5_10 + @noedge6_10 + 2 d + @obj + ;

rup >= 1 ;

output NONE ;

conclusion BOUNDS 8 8 ;

end pseudo-Boolean proof ;

Three colour classes here, so three at-most-one constraints, all added to the solution-improving constraint

Verifying This Proof (For Real, This Time)

```
$ veripb clique.opb clique-new-attempt.pbp
Running VeriPB version 3.0.2
s VERIFIED BOUNDS 8 <= obj <= 8
```

No errors this time — every step is justified

Note that the checker **knows nothing** about

- cliques,
- colourings, or
- bound functions

Verifying This Proof (For Real, This Time)

```
$ veripb --trace clique.opb clique-new-attempt.pbp
Running VeriPB version 3.0.2
Objective: min 1 ~x1 1 ~x2 . . . 1 ~x11 1 ~x12 0
ConstraintId 1: 1 ~x1 1 ~x3 >= 1
. . .
ConstraintId 41: 1 ~x11 1 ~x12 >= 1
line 2: @obj soli x7 x9 x12 ;
ConstraintId 42: 1 x1 1 x2 . . . 1 x11 1 x12 >= 4
line 3: rup 1 ~x12 1 ~x7 >= 1 ;
ConstraintId 43: 1 ~x7 1 ~x12 >= 1
line 4: pol @noedge1_6 @noedge1_9 + @noedge6_9 + 2 d @obj + ;
ConstraintId 44: 1 x2 1 x3 1 x4 1 x5 1 x7 1 x8 1 x10 1 x11 1 x12 >= 3
. . .
line 16: rup >= 1 ;
ConstraintId 55: >= 1
s VERIFIED BOUNDS 8 <= obj <= 8
```

Why a General-Purpose Proof System?

Why not just put a “colouring bound” rule into the proof format?

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- And we would need to do the same again for every other kind of solver

We would rather not have a proof format and a proof checker per solver, per paper...

Cutting planes turns out to be strong enough for all of these cases

And the derivation length scales as the solver reasoning time (asymptotically)

Different Clique Algorithms

Different search orders?

- ✓ Irrelevant for proof logging

Using local search to initialise?

- ✓ Just log the incumbent

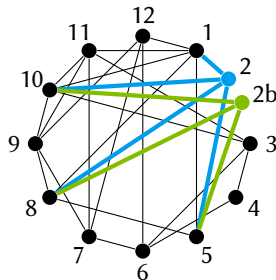
Different bound functions?

- Is cutting planes strong enough to justify every useful bound function ever invented?
- So far, seems like it...

Weighted cliques?

- ✓ Multiply a colour class by its largest weight
- ✓ Also works for vertices “split between colour classes”

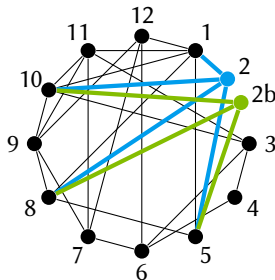
Lazy Global Domination for Maximum Clique [MP16]



Can ignore vertex 2b

- Every neighbour of 2b is also a neighbour of 2
- This is an example of **dominance** reasoning

Lazy Global Domination for Maximum Clique [MP16]



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Dominance rule can justify this

- Even when detected dynamically during search

Reducing the Trust Base

We are still trusting rather a lot

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- VERIPB can **elaborate** a proof into a simpler **kernel format**
- Uses only rules that the **formally verified** checker **CAKEPB** knows about
- CAKEPB (written in CAKEML and proved correct in HOL4)
 - reads the **original graph file**
 - does a formally verified translation to 0–1 ILP
 - checks the kernel proof against this translation

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- So the answer is verified in terms of the original input

End-to-End Verification for Maximum Clique

```
$ glasgow_clique_solver brock200_4.clq
recursions = 56613
clique = 17 (12 19 28 29 38 54 65 71 79 93 117 127 139 161 165 186 192)
runtime = 0.02153s
```

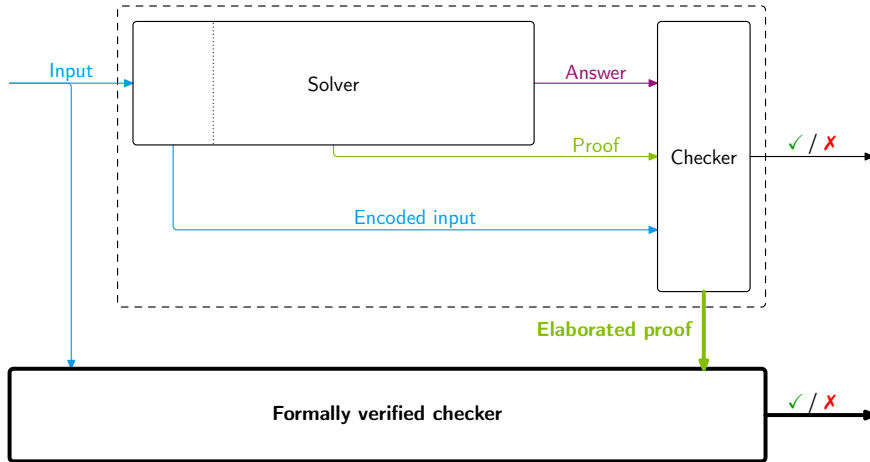
```
$ glasgow_clique_solver brock200_4.clq --prove brock200_4
runtime = 0.266475s
```

```
$ cake_pb_clique brock200_4.clq > brock200_4.verifiedopb
```

```
$ veripb --elaborate brock200_4.corepb brock200_4.verifiedopb brock200_4.pbp
s VERIFIED BOUNDS 183 <= obj <= 183
```

```
$ cake_pb_clique brock200_4.clq brock200_4.corepb
s VERIFIED MAX CLIQUE SIZE |CLIQUE| = 17
```

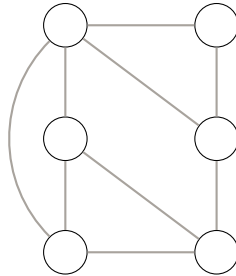
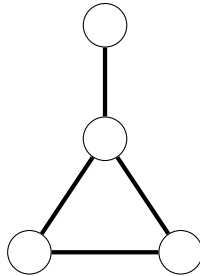
Workflow for Solving with End-to-End Certified Results



Still have to trust

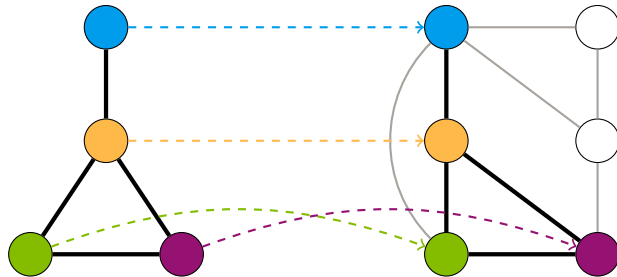
- The HOL4 theorem prover
- That the formal model of the CAKEML environment matches the hardware it runs on
- The HOL4 definition of what it means to be a maximum clique
- Input parsing and output formatting

Subgraph Isomorphism



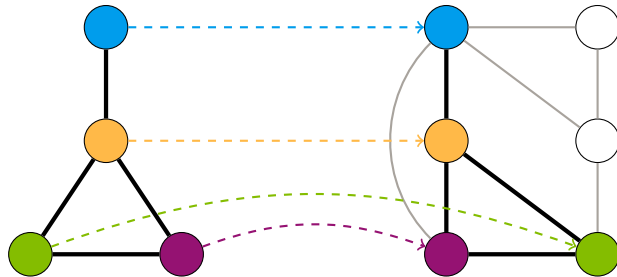
- Find the **pattern** inside the **target**
- Applications in compilers, biochemistry, model checking, pattern recognition, ...
- Often want to find **all** matches

Subgraph Isomorphism



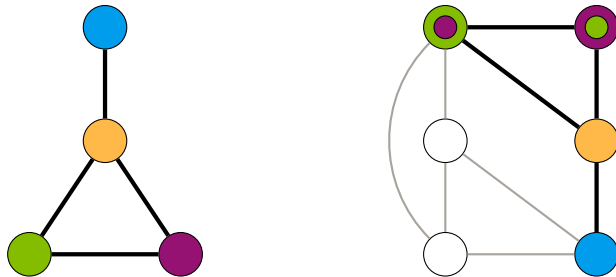
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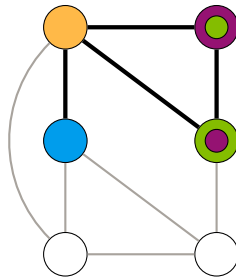
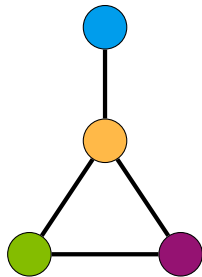
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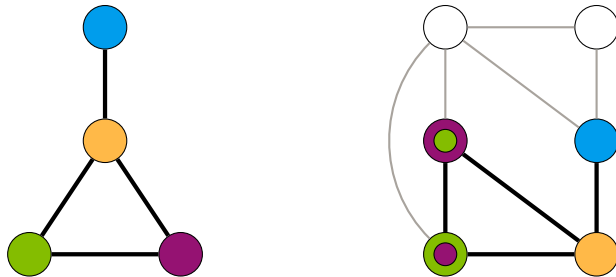
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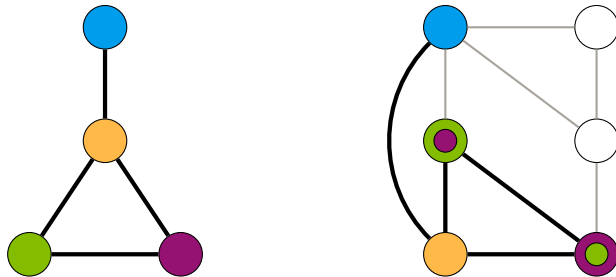
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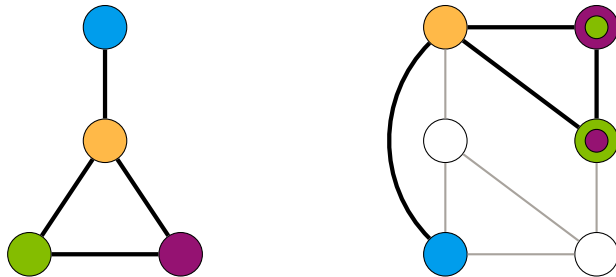
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Subgraph Isomorphism in Pseudo-Boolean Form

Each pattern vertex gets a target vertex:

$$\sum_{t \in V(T)} x_{p,t} = 1 \qquad p \in V(P)$$

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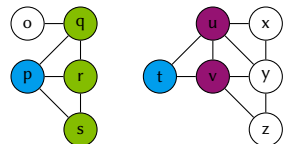
Each target vertex may be used at most once:

$$\sum_{p \in V(P)} -x_{p,t} \geq -1 \quad t \in V(T)$$

Adjacency constraints: if p is mapped to t , then neighbours of p must be mapped to neighbours of t :

$$\bar{x}_{p,t} + \sum_{u \in N(t)} x_{q,u} \geq 1 \quad p \in V(P), q \in N(p), t \in V(T)$$

Degree Reasoning in Cutting Planes

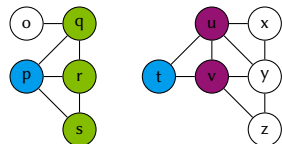


Pattern vertex p of degree $\deg(p)$ can never be mapped to target vertex t of degree $< \deg(p)$

Observe $N(p) = \{q, r, s\}$ and $N(t) = \{u, v\}$

We wish to derive $\bar{x}_{p,t} \geq 1$

Degree Reasoning in Cutting Planes



Adjacency:

$$\bar{x}_{p,t} + x_{q,u} + x_{q,v} \geq 1$$

$$\bar{x}_{p,t} + x_{r,u} + x_{r,v} \geq 1$$

$$\bar{x}_{p,t} + x_{s,u} + x_{s,v} \geq 1$$

Injectivity:

$$-x_{o,u} + -x_{p,u} + -x_{q,u} + -x_{r,u} + -x_{s,u} \geq -1$$

$$-x_{o,v} + -x_{p,v} + -x_{q,v} + -x_{r,v} + -x_{s,v} \geq -1$$

Literal axioms:

$$x_{o,\textcolor{violet}{u}} \geq 0$$

$$x_{o,v} \geq 0$$

$$x_{p,u} \geq 0$$

$$x_{p,v} \geq 0$$

Add these together and divide by 3 to get

$$\bar{x}_{p,t} \geq 1$$

Possible to certify (almost) all of the other things state-of-the-art subgraph solvers do:

- Conditional injectivity reasoning and filtering
- Distance filtering
- Neighbourhood degree sequences
- Path filtering
- Supplemental graphs

Limitations

Why trust the encoding?

- ✓ For clique and subgraph isomorphism this is now taken care of [GMM⁺24]
 - Has to be done again for each new problem, but not too bad

Constraints can be specified [extensionally](#)

Variable assignments must match some row in table

define

$$(A, B, C) \in [(1, 2, 3), (1, 3, 4), (2, 2, 5)]$$

$$\text{i.e., } t_1 \Rightarrow (a_{=1} \wedge b_{=2} \wedge c_{=3})$$

$$\text{i.e., } t_2 \Rightarrow (a_{=1} \wedge b_{=4} \wedge c_{=4})$$

$$\text{i.e., } t_3 \Rightarrow (a_{=2} \wedge b_{=2} \wedge c_{=5})$$

using newly introduced tuple selector variables

$$t_1 + t_2 + t_3 = 1$$

Tuple variables **used for proof logging** — need not be known by solver (except for printing proofs)

Restarts are easy

No need to justify guesses or decisions — only justify backtracking

$$Z \in \{1, 3\}$$

$$\begin{array}{l} V \in \{1 \quad 4 \quad 5\} \\ W \in \{1 \quad 2 \quad 3\} \\ X \in \{ \quad 2 \quad 3\} \\ Y \in \{1 \quad 3\} \\ Z \in \{1 \quad 3\} \end{array} \quad w_{=1} + w_{=2} + w_{=3} \geq 1 \quad [W \text{ takes some value}]$$

$V \in \{1 \quad 4 \quad 5\}$				
$W \in \{1 \quad 2 \quad 3 \quad \}$	$w_{=1} +$	$w_{=2} +$	$w_{=3}$	≥ 1 [W takes some value]
$X \in \{ \quad 2 \quad 3 \quad \}$		$x_{=2} +$	$x_{=3}$	≥ 1 [X takes some value]
$Y \in \{1 \quad 3 \quad \}$	$y_{=1}$	$+$	$y_{=3}$	≥ 1 [Y takes some value]
$Z \in \{1 \quad 3 \quad \}$	$z_{=1}$	$+$	$z_{=3}$	≥ 1 [Z takes some value]

ACP Summer School '26 120/126

How do we know we're giving a proof for the problem the user actually specified?

Need modelling language with formally specified semantics...

The *Glasgow* Constraint Solver

Support for 37 constraints (and counting), with all inferences certified:

Abs, AllDifferent, AllEqual, Among, AtMostOne, BinPacking, Circuit, Comparison, Count, Cumulative, Diffn, Disjunctive, DivMod, Element, Equals, GCC, In, Increasing, Inverse, Knapsack, Lex, Linear, Logical, MDD, MinDistance, MinMax, Multiply, NValue, Parity, PlusMinus, Power, Regular, SeqPrecedeChain, SmartTable, Sort, Table, ValuePrecede

Also **certifies reformulations** not written in but inferred from input:

- Auto-tabulation: Identify small subproblem, solve it, and replace by table

Performance of pseudo-Boolean proof logging and checking

- Faster proof logging and checking!
- Trim proofs while checking as in SAT solving [HHW13a] (*upcoming work [ABB⁺26a]*)
- Compress proof files using binary format

- Model counting
- Mixed integer linear programming (*suggested extension of VERIPB in [DEGH23]*)
- Satisfiability modulo theories (SMT) solving (*Lots of work in, e.g., [HS22, BBC⁺23, RSB⁺26, RSS⁺26]*)

- Use proof logs for algorithm analysis or explainability purposes
- Lots of other challenging problems and interesting ideas





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